Delegation cascades: why safe instructions produce unsafe agents

Author One[1] and Author Two[1,2]

[1]Affiliation One
[2]Affiliation Two

Abstract

We study what happens to a safety instruction when it is passed down a chain of delegated agents before anything is done with it. A principal enters a competitive race but does not act in it: it issues a specification to an agent, which issues one to a sub-agent, and so on for d hand-offs. Two things happen along the chain, and both are geometric in d. The specification is transmitted imperfectly, so it survives d hand-offs intact with probability $(1-\varepsilon)^d$. Responsibility is traced back only in part, so a principal at depth d is charged the fraction ϕ^d of the harm it causes. We embed both in the reduced four-strategy AI race of Fernández Domingos and Han, make delegation depth part of the strategy, and analyse the resulting dynamics. Five results follow. First, depth is an unbounded liability shelter. Realised harm saturates while attributed harm decays, so for every $\phi < 1$ there is a depth beyond which each further hand-off strictly lowers the bill even though it strictly raises the damage. No level of liability stops chains from lengthening past it. Second, and against the intuition that the two mechanisms simply add up, neither is dangerous on its own. Start from a race that liability has already made safe at depth zero. Imperfect transmission alone lifts the long-run unsafe frequency to 0.018 and imperfect attribution alone to 0.012; together they lift it to 0.322, so 91% of the joint effect is an interaction. A counterfactual that freezes the population composition isolates its single cause. Under full pass-through the market answers transmission loss by *shortening* its chains, from six hand-offs to two, and per-layer attribution is precisely what disables that answer. Third, the resulting equilibrium is one in which every principal issues the safest available instruction and the population executes the unsafe designs anyway: declared and executed behaviour differ by 0.74 in total variation. Fourth, instruments that act on the incentive to delegate behave monotonically while instruments that act on transmission fidelity do not. A partial fidelity improvement can leave the ecosystem worse off than no improvement at all, because over-specification was an equilibrium response to erosion. A ceiling of one hand-off removes the unsafe behaviour entirely and raises social payoff from -0.8 to 60.5. Fifth, the shelter turns on attribution vanishing in depth and not on its geometric form: splitting the blame equally across the chain reproduces it exactly, while any rule bounded away from zero closes it. A statutory floor under attribution traces the same safety and welfare frontier as a hard cap on chain length, which matters because a floor asks no regulator to count how many hands an instruction passed through.

Keywords: evolutionary game theory; delegation; principal-agent chains; liability; AI governance; multi-agent systems

1 Introduction

Software that acts on behalf of someone else is increasingly built as a chain. A person instructs an agent, the agent decomposes the task and instructs sub-agents, and the sub-agents call further tools and agents. The process that finally takes an action in the world is then several hand-offs away from the party that wanted it taken [1, 2]. Governance of such systems is written almost entirely at the two ends of the chain. Safety training and evaluation act on the model; liability, procurement rules and disclosure requirements act on the principal [3, 4]. The hand-offs in between are treated as plumbing.[1]

They are not plumbing. A hand-off does two things that matter. It transmits a specification imperfectly, because each layer restates the task in its own terms and the part most easily dropped is the qualifying clause rather than the task itself. Degradation under repeated re-encoding is a robust property of transmission chains, in human cultural transmission, where the distinctions that nothing selects for are the first to be lost [6], in recursive machine generation [7, 8] and in chains of language models that each restate what the previous one wrote, with a systematic bias in which content survives [9, 10]. The same decay has been measured on instructions directly: across fourteen frontier models, adherence to explicitly stated instructions falls with every additional turn, from 0.88 to 0.71 over three turns for the strongest of them [11]. That setting is the easiest possible case, one model with the whole history in front of it and nothing restated by a third party, so it bounds the erosion of a specification from below rather than measuring it. Failure data from deployed agent frameworks show the same two facts in the field: specification errors between agents are the dominant failure class, and a linear chain of hand-offs degrades more than any other topology [12, 13]. And it attenuates responsibility, because harm that reaches the world through several intermediaries is traced back to the party at the top only in part. The gap this opens has been named for the case of a machine whose behaviour its operator cannot predict [14]; ours is a different source of the same gap, structural rather than epistemic, and one a principal can widen at will by adding a layer. Both effects are per-layer, so both compound geometrically with the number of hand-offs, and a cost that compounds in that way is the classical reason a long hierarchy is expected not to survive competition [15]. What is not obvious is what a population of principals does about them.

The question matters because the two effects push the principal in opposite directions. Imperfect transmission is a private cost: a principal who is held responsible for what its chain does has every reason to keep the chain short. Imperfect attribution is a private benefit: depth buys a discount on the bill. Which force wins is not a matter of taste. Neither is the answer to the policy question that follows: whether an instrument that improves transmission, one that restores attribution, or one that simply caps the length of the chain does more for the same effort.

We answer both by making delegation depth a strategic variable in an evolutionary model. The interaction layer is the reduced four-strategy AI race of Fernández Domingos and Han [16], used without modification, so that every depth effect we report is attributable to the delegation layer we build on top of it. A design is a pair (d, s): a number of hand-offs and the specification the principal issues. One hand-off acts on the specification by a row-stochastic kernel, and the

[1] We call the result a delegation cascade. The term denotes compounding per-hand-off loss along a single chain from a principal to the party that acts, and not the informational cascade of Bikhchandani et al. [5], in which independent decision-makers rationally discard their own signals after observing what their predecessors did.

harm a chain causes is charged to its principal at the rate ϕ^d. Principals do not optimise; designs that pay better are copied more, in the standard replicator and finite-population dynamics of evolutionary game theory [17–19].

Our contribution is an explanation, not a method or a benchmark. In one sentence: *delegation depth is a strategic variable whose two effects are individually minor and jointly decisive, because per-layer attribution disables the market's own response to specification erosion.* The results below make that precise.

Results. We show, first, that depth is an unbounded liability shelter (Theorem 4). Realised harm saturates in d because the specification is eventually forgotten, while attributed harm carries the extra factor ϕ per layer. For every $\phi < 1$ there is therefore a depth beyond which each further hand-off strictly reduces the harm the principal is charged for while strictly increasing the harm it causes. Beyond that depth the invasion condition for one more layer stops depending on liability in the usual direction: raising the penalty makes the deeper design *more* attractive, not less (Proposition 5).

Second, the two mechanisms do not add. We calibrate the liability so that the depth-zero race is already safe: the effective liability we use is 3.1 times the level at which the single-layer game has long-run unsafe frequency zero. At that setting, imperfect transmission alone yields 0.018 and imperfect attribution alone yields 0.012, while the two together yield 0.322. The interaction, +0.293, is 91% of the joint effect. A counterfactual that holds the population composition fixed shows why. Under full pass-through the population answers transmission loss by shortening its chains from six hand-offs to two. That answer, and not any change in behaviour at a given depth, is what per-layer attribution removes.

Third, the equilibrium that results is one in which safety declarations are uninformative. Every principal issues AS, the safest available specification, and the population executes AS only 26% of the time; the distance between what is declared and what is executed is 0.74 in total variation. Over-specification is the equilibrium response to erosion, and it does not repair it.

Fourth, the four governance instruments we compare are not interchangeable. Restoring attribution has a threshold rather than a gradient: nothing happens until ϕ reaches 0.86, after which chains collapse from six hand-offs to two. Capping chain length is monotone and cheap, and a ceiling of one hand-off both removes the unsafe behaviour and raises social payoff from −0.8 to 60.5. Improving transmission fidelity is neither monotone nor safe in the small. Taking the erosion probability from 0.20 to 0.07 reduces the unsafe frequency to 0.032, but taking it to 0.04 raises it again to 0.239, because the improvement removes the reason principals were over-specifying. Where a single check is placed in the chain matters by a factor of 18, and it should be placed at the agent that acts.

Assumptions. Three should be stated at the outset rather than at the end. The chain transmits one specification per race, so the executed designs of the two seats are independent draws and every payoff is bilinear; a model in which layers re-negotiate during play would not have this structure. The erosion kernel is exogenous and is not itself under selection, so we describe what a market does under a given transmission technology, not how that technology evolves. And attribution decays geometrically, which is a modelling choice that we treat as the definition of per-layer liability rather than as an empirical claim; Section 6 reports what changes when it is replaced.

2 Related work

Evolutionary models of AI races. That a development race trades safety for speed, and that the trade worsens as the field gets more crowded, is the argument of Armstrong et al. [20]; treating that race as an evolutionary game played in a population of developers is due to Han et al. [21]. The reduced four-strategy race we use as our interaction layer is the model of Fernández Domingos and Han [16], in which participants repeatedly choose Safe or Unsafe, unsafe development pays more per round and advances faster, and a leader who cut corners carries a private risk of a setback. That line of work has since examined the choice between sanctioning unsafe development and rewarding safe development [22], heterogeneous and networked populations [23], voluntary safety commitments [24], the incentives of the auditors a regulator would have to rely on [25], and the trust of the users who face the result [26, 27], all as ways of moving the equilibrium; outside this lineage the same trade has been modelled as a collective action problem [28]. All of it places the strategic agent in the race. Our contribution is orthogonal: we leave the race exactly as it is and ask what happens when the party whose design is being selected is several hand-offs removed from the party that acts.

Delegation to artificial agents. Behavioural work has shown that delegating a decision to an artificial agent changes what people choose, sometimes towards more prosocial outcomes [29], and that people who program an artificial delegate transmit their intent accurately but not precisely [30]; when trusting a machine is warranted at all has been asked in the same evolutionary terms [31]. That literature studies a single delegation step, from a human to a machine. Accountability degrades because of the structure of an algorithmic supply chain rather than the conduct of any single actor in it, and visibility along such a chain ends at what has been called the accountability horizon; both are established arguments in the governance literature [32]. The accountability problems raised specifically by multi-step agentic delegation have begun to be described [33–35], the liability questions raised by chains of LLM-based sub-agents have been catalogued in principal-agent terms [36], and legal scholarship has started to read the relation between a principal and the AI agent acting for it through the doctrines built for human agents [37, 38], but as far as we are aware without a formal model in which the depth of the chain is itself selected. Making depth a strategic variable is what allows the two mechanisms to be separated and their interaction to be measured.

Hierarchies and the loss of control. That authority degrades as it passes down a hierarchy is a classical observation in the economics of organisation. It appears as the loss of control that bounds firm size [39, 40], as the cost of decentralised information processing [41], as collusion between layers [42], as the wedge between the authority a principal formally holds and the authority an informed subordinate actually exercises [43], and as the question of how many tiers to build at all when every tier costs control [15, 44, 45]. Contract theory asks in the same spirit what is lost when the contract with the party that acts is written one tier down rather than by the principal itself [46, 47], and in a benchmark model with adverse selection and collusion, delegating to one agent the right to subcontract with the next earns the principal strictly less than contracting centrally [48]. Among the conditions under which that loss is mitigated entirely is one our architecture denies: the principal never observes what one of its agents passes to the next.

Diluted responsibility and the judgment-proof problem. Diluted responsibility in teams is equally classical [49], and the difficulty of tracing an outcome back to any one contributor when many hands have produced it was named as the first barrier to accountability in computerised systems three decades ago [50]. Experiments find that principals delegate precisely to shed blame: an unfair choice routed through an agent draws less punishment, and is made more often, than the same choice made directly [51, 52]. The law has long recognised that the choice of liability rule shapes how much care is taken [53] and that liability must sometimes be assigned to a party other than the actor for incentives to survive an intermediary [54]. Where that assignment stops, at the edge of the scope of employment and at the doctrines exempting a principal from the torts of an independent contractor, has been analysed in the same economic terms [55], and it is that boundary, crossed once per hand-off, that our attribution rate encodes. Liability can be assigned by making a third party the gatekeeper of the harmful act [56] or by reading an artificial agent as the servant of whoever put it to work [57]; how those assignments extend to AI systems, and where the European liability instruments still leave the bill unassigned, is by now a literature of its own [58, 59]. The AI Liability Directive those critiques address was withdrawn in October 2025 [60], leaving the AI Act and the revised Product Liability Directive as the operative instruments. Closest to the mechanism we report is the judgment-proof problem [61]: an injurer whose assets fall short of the harm it can cause takes too little care, because the part of the bill it will never be presented with does not enter its calculation. That status can be engineered rather than suffered, by structuring obligations so that little is left to seize [62], and liability extended to deeper pockets can be undone by the same restructuring [63]. The response has been measured in one setting: as United States courts extended liability for long-latency occupational harms, entry of small corporations into the hazardous sectors rose sharply, which Ringleb and Wiggins [64] read as firms divesting the risky stage into entities that would not be able to pay. Depth is a way of becoming judgment proof by construction rather than by insolvency, and unlike a balance sheet it has no natural bound. Our model brings the two together and adds the ingredient those literatures do not have: a population in which the depth of the hierarchy is not chosen by an optimising designer but selected by differential replication of designs. Putting a property other than strategy under selection alongside it is the subject of coevolutionary games, where that property is typically the interaction network, reputation, mobility or age [65], and the closest precedent for a structural property of the kind we study is the evolution of hierarchy in human groups [66]. To our knowledge the number of intermediaries between a strategy and its execution has not been placed under selection before.

Specification and alignment. The gap between a stated objective and an executed one is the subject of the specification-gaming literature [67], and incomplete contracting has been proposed as the right frame for alignment [68]. Those accounts locate the gap between the principal and a single agent. We locate a second gap, which compounds with the number of intermediaries and which no amount of care at either end of the chain removes.

Protocols for traceability and runtime oversight. A parallel and mostly separate literature builds the machinery that would make a delegation chain accountable in practice: agent identifiers, activity logs and the rest of the infrastructure that would let an action be traced back to the party that commissioned it [4, 69, 70], together with proposed operating practices for the parties in an agentic deployment [3]; delegation itself can be authenticated, each hand-off carrying a verifiable credential from the party above it [71]. The Agent Control Protocol [72] places a cryptographic admission check in front of every agent action, carries capabilities along a verifiable chain of delegations, and writes each decision to a tamper-evident ledger from which the authorisation chain can be reconstructed; an action is authorised by a single-use execution token issued only after the check passes. He et al. [73] take the runtime rather than the protocol route, modelling a multi-agent system as a dynamic execution graph and attaching an oversight agent that detects anomalies at the level of nodes, edges and paths and can intervene during execution. Both lines are engineering answers to questions our model poses in economic terms, and Section 5 sets out the correspondence. What neither literature asks, and what we ask here, is what a population of principals chooses to build when such machinery is absent, partial or merely optional: these are mechanisms that change ϕ, the placement of a check and the shape of the erosion kernel, and the equilibrium consequences of moving those are the subject of this paper.

Revision timing. We treat the evolutionary layer as slow relative to a race and take revision opportunities to arrive as a Poisson process, the standard assumption behind the pairwise-comparison dynamics we use. Kara et al. [74] show that when a strategy is decomposed into sub-tasks the inter-revision intervals become Erlang rather than exponential and the resulting population dynamics are not those of the memoryless case. A delegation chain is exactly such a decomposition, so the timing assumption is not innocuous; we state it explicitly in Section 3 and return to it in Section 6.

3 Model

3.1 The interaction layer

Two seats play the repeated race of Fernández Domingos and Han [16]. In each round a seat plays Safe or Unsafe. Unsafe pays more in the stage game, with round payoffs 1.0 and 0.6 for a safe player against a safe and an unsafe opponent against 2.4 and 2.0 for an unsafe one, and it contributes 1.5 rather than 1 to race progress. The race lasts T rounds, with T guaranteed to be at least five and stopping with probability 0.2 per round thereafter, so $\mathbb{E}[T] = 9$. The leader in accumulated progress at the horizon takes a prize of 100, split on a tie, and a seat that is at risk loses everything it has accumulated with probability $p_{\max} n^U/T$, where n^U is its number of unsafe rounds and $p_{\max} = 0.6$.

Following the source study we work with four reduced designs, written here in the order in which their specifications lose clauses: AS is unconditionally safe, CS opens Safe and then copies the opponent, CAS opens Unsafe and then copies, and AU is unconditionally unsafe. All four are deterministic, so the action path of an ordered pair is fixed once the pair is fixed and the only stochastic primitive is the horizon. We therefore evaluate every matchup by exact expectation over the horizon law rather than by sampling, which removes simulation noise from the payoff matrix entirely. Write $a(i, j)$ for the expected task payoff of design i against j, $m(i, j)$ for its expected number of Unsafe actions and $u(i, j)$ for its expected fraction of Unsafe rounds.

Reading down any column of m in Table 1, the entries never fall. This monotonicity is the structural fact on which everything below rests, and we record it as a property of the interaction layer rather than an assumption of ours. It holds round by round and not merely in expectation,

Table 1: Interaction layer at depth zero. Task payoff $a(i,j)$ and expected number of Unsafe actions $m(i,j)$ of the focal design i against j, evaluated exactly over the horizon law. Reading down a column, m never falls: the erosion order is a harm order.

	$a(i,j)$				$m(i,j)$			
i	AS	CS	CAS	AU	AS	CS	CAS	AU
AS	59.00	59.00	8.60	5.40	0.00	0.00	0.00	0.00
CS	59.00	59.00	26.64	16.60	0.00	0.00	4.22	8.00
CAS	101.77	61.63	27.20	27.20	1.00	4.78	9.00	9.00
AU	48.64	47.36	27.20	27.20	9.00	9.00	9.00	9.00

Table 2: The two mechanisms, separately and together. Long-run Unsafe frequency U, mean delegation depth $\bar{d}$, declaration gap and social payoff in the stationary regime of the finite-population process. Neither mechanism does much alone; the interaction is 0.293, or 91% of the joint effect.

erosion ε	attribution ϕ	U	$\bar{d}$	gap	π_S
0	1	0.0000	6.00	0.000	68.0
0.2	1	0.0176	1.99	0.358	58.3
0	0.75	0.0115	5.95	0.000	65.5
0.2	0.75	0.3225	6.00	0.738	-0.8
interaction		+0.2934			

Table 3: Depth ceilings. Long-run Unsafe frequency and social payoff when chains longer than $\bar{D}$ hand-offs are forbidden. The social payoff is maximised well inside the ceiling the market selects.

$\bar{D}$	U	$\bar{d}$	π_S
0	0.0000	0.00	59.0
1	0.0000	1.00	60.5
2	0.0178	2.00	58.3
3	0.0640	3.00	49.9
4	0.1359	4.00	36.0
5	0.2251	5.00	18.5
6	0.3225	6.00	-0.8

which is what makes it independent of every numerical choice in the paragraph above.

Lemma 1 (The erosion order is a harm order)**.** *Write* $a_t^{i|j} \in \{\mathrm{S}, \mathrm{U}\}$ *for the action taken in round* t *by design* i *against design* j*, and order the two actions* $\mathrm{S} < \mathrm{U}$*. For every opponent* j*, every horizon* n *and every round* $t \le n$*,*

$$a_t^{\mathsf{AS}|j} \;\le\; a_t^{\mathsf{CS}|j} \;\le\; a_t^{\mathsf{CAS}|j} \;\le\; a_t^{\mathsf{AU}|j}. \tag{1}$$

Consequently $m(\mathsf{AS}, j) \le m(\mathsf{CS}, j) \le m(\mathsf{CAS}, j) \le m(\mathsf{AU}, j)$ *and the same holds for* u*, for every horizon law and every setting of the stage payoffs, the step sizes, the prize,* $\mathrm{p}_{\max}$ *and the setback scope.*

Proof. Write a design as a pair (f, c): a first action f and a flag c for whether it copies its opponent's previous action. Solving the two coupled recursions gives the focal path in closed form in each of three cases. If the opponent does not copy it plays f_j in every round whatever the focal seat does, so a focal seat that does not copy plays f_i throughout, and one that copies plays f_i in round 1 and f_j afterwards. If both copy then $a_t = a_{t-2}$ for $t \ge 3$, so the focal seat plays f_i in odd rounds and f_j in even ones. In every case the path is either constant at f_i, or equal to f_i on a set of rounds that does not itself depend on f_i and to f_j on the complement.

AS plays Safe in every round and AU plays Unsafe in every round, against every opponent, so they are the pointwise smallest and largest paths in the design set and the outer two inequalities of (1) hold. CS and CAS carry the same flag $c = 1$ and differ only in f_i, which enters each closed form monotonically, so raising f_i from Safe to Unsafe raises the path in some rounds and lowers it in none. This gives the middle inequality.

At a fixed horizon m counts the Unsafe entries of the path and u divides that count by n, so both are non-decreasing functions of the path and both inherit (1). Neither depends on any other parameter of the race, so taking the expectation over any law for T preserves the ordering. □

The harm matrix in closed form. The same case analysis evaluates m exactly rather than only ordering it. With $\mu = \mathbb{E}[T]$ and $\pi = \Pr[T \text{ odd}]$, the rows of m in the erosion order are $(0, 0, 0, 0)$ for AS, $(0, 0, \frac{\mu-\pi}{2}, \mu - 1)$ for CS, $(1, \frac{\mu+\pi}{2}, \mu, \mu)$ for CAS and (μ, μ, μ, μ) for AU. At our horizon law $\mu = 9$ and $\pi = 5/9$, which reproduces Table 1 exactly. Every entry is bounded by μ, which is the saturation Theorem 4 needs.

3.2 The delegation chain

A seat is operated not by a principal but by a chain

$$\text{principal} \;\longrightarrow\; \text{agent}_1 \;\longrightarrow\; \cdots \;\longrightarrow\; \text{agent}_d \;\longrightarrow\; \text{the race},$$

where $d \ge 0$ is the *delegation depth*. The principal issues one specification, its *intent* s, and each hand-off transmits it through the row-stochastic kernel

$$M \;=\; (1-\varepsilon)\, I \;+\; \varepsilon\, D, \tag{2}$$

with ε the per-hand-off erosion probability and D a drift kernel on the four designs. The identity term in (2) is the part of a specification that survives a hand-off untouched. Our baseline D is the *ladder*: an erosion event moves the specification one rung down the order AS, CS, CAS, AU, which is to say it drops one safety clause and leaves the description of the task intact. Two

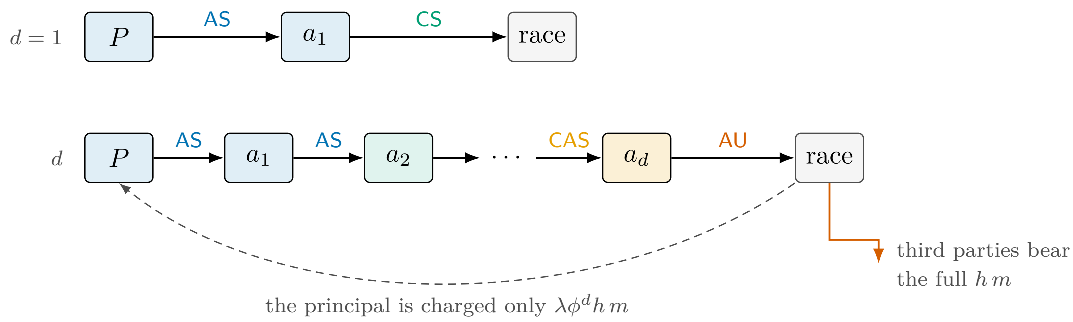

Figure 1: A delegation cascade. A principal P issues one specification and never acts. Each hand-off may drop a safety clause, so the design that reaches the race is a random function of the intent and of the depth alone; the deeper the chain, the further the executed design has slid down the erosion order AS $\to$ CS $\to$ CAS $\to$ AU. The harm the chain causes falls on third parties in full, but is charged back to the principal at the rate ϕ^d. Both effects are per-layer, so both are geometric in the depth, and they pull the principal in opposite directions: erosion makes depth expensive, attenuated attribution makes it cheap.

alternatives are used as controls. Under *collapse* an erosion event discards the specification and the sub-agent falls back on the locally competitive design AU. Under *uniform* it replaces the specification with a uniformly random one; that kernel has the same spectral gap [75] as the ladder but no direction, and it separates the effect of biased erosion from the effect of noise as such.

The executed design of a depth-d chain with intent s is therefore distributed as row s of M^d, written $p_{d,s}$. Figure 1 shows the construction, and Figure 2 shows what it does to an AS instruction.

We also allow one *check* in the chain. A check of strength α at layer $k \le d$ restores the intent with probability α, after which the specification still has $d-k$ hand-offs to survive, so the executed design is distributed as $\alpha M^{d-k}(s,\cdot) + (1-\alpha)M^d(s,\cdot)$.

3.3 Three functionals

Two designs meeting in the race produce, in expectation over the executed designs and over the horizon,

$$a(d,s;d',s') = p_{d,s}^\top A\, p_{d',s'} + B(d), \qquad m(d,s;d',s') = p_{d,s}^\top M_h\, p_{d',s'}, \tag{3}$$

where A and M_h are the task-payoff and unsafe-action matrices of the interaction layer and $B(d) = g\,d - c\,d^2$ is the net organisational benefit of running a chain of depth d. Everything is bilinear because one specification is transmitted per race and then played, so the executed designs of the two seats are independent draws. The benefit B is real: delegation genuinely produces more, and the term enters the social accounting as well as the private one. The baseline sets $g = 1.5$, about 2.5% of the payoff of a safe seat per hand-off, and $c = 0$, so that in the baseline the only force limiting depth is the harm the principal internalises.

Let h be the external harm of one Unsafe action and λ the liability rate at depth zero. The *private* functional, which alone drives the dynamics, and the *social* functional, which alone scores them, are

$$\pi_P(d,s;d',s') = a(d,s;d',s') - \lambda\,\phi^d\,h\;m(d,s;d',s'), \tag{4}$$

$$\pi_S(d,s;d',s') = a(d,s;d',s') - h\;m(d,s;d',s'), \tag{5}$$

The two functionals in (5) differ only in the second term, and their difference is the *dilution wedge*

$$\Delta(d) \;=\; \pi_P - \pi_S \;=\; h\big(1 - \lambda\phi^d\big)\, m, \tag{6}$$

The wedge (6) is non-decreasing in d for every $\phi \leq 1$ and vanishes identically at $\phi = \lambda = 1$. Depth alone widens the gap between what a principal optimises and what society bears, without any change in what the principal intends. Only the product $L = \lambda h$ enters π_P, so the two-parameter family (λ, h) collapses to the single control L, the *effective liability*; h is a property of the world and stays fixed when the instrument moves, which is what keeps social payoffs comparable across liability regimes. Keeping the two functionals apart is not bookkeeping. Once the cost of an intervention is counted, driving a behavioural outcome to its maximum and maximising total population payoff are different objectives, and it is the second an intervention should be designed against [76]; we therefore report the long-run unsafe frequency and the social payoff separately throughout, and never combine them.

3.4 Dynamics and the choice of parameters

Three timescales are separated, and the separation is an assumption rather than a derivation. A race runs to its horizon, $\mathbb{E}[T] = 9$ rounds, on the fastest one; a chain is built once per race and transmits one specification, with no renegotiation while the race is under way; and designs are revised on the slowest, so that a principal reconsiders how deeply to delegate only between races and never inside one. The separation is what makes the executed designs of the two seats independent draws and the functionals bilinear. It is also what a model with explicit revision clocks would relax, and we return to this in Section 6.

Designs are copied in proportion to how well they pay. We report the stationary regime of the finite-population pairwise-comparison process [77] in the small-mutation limit [78–80] as the primary reading, with population size $Z = 100$ and selection intensity $\beta = 0.05$, and the basin-averaged attractor of the replicator equation as a secondary one. The joint design space has $4(D+1)$ elements, 28 at the baseline ceiling $D = 6$, which puts the full mutation-selection chain out of reach: its state space has $\binom{Z+n-1}{n-1}$ elements. We therefore compute fixation probabilities in closed form, checked against the general-purpose implementation of Fernández Domingos et al. [81] on games small enough for it, and check the reduction against the full chain on a reduced design space (Section 4.8), where the two agree to 0.012 at the smallest mutation rate examined. How small a mutation rate the embedded chain needs for that agreement to hold has been characterised analytically for imitation processes at arbitrary selection intensity [82], which is the class our update rule belongs to.

The remaining choice is the liability level, and we fix it so that the delegation results cannot be an artefact of a race that was unsafe to begin with. Bisection on the depth-zero game puts the long-run unsafe frequency below 10^{-3} for every $L \geq 6.52$; we run at $L = 20$, that is 3.1 times the level at which the single-layer race is already safe. At that level the depth-zero ecosystem has long-run unsafe frequency 0.000 under the finite-population reading and 0.026 under the replicator reading. The baseline erosion probability is $\varepsilon = 0.20$ and the baseline attribution retention is $\phi = 0.75$; both are swept over their full ranges in Section 4.4 and Section 4.8.

4 Results

4.1 The specification is forgotten geometrically

Under the ladder kernel the number of clauses lost over d hand-offs is binomial, so the executed design is the intent shifted down the erosion order by $\mathrm{Bin}(d, \varepsilon)$ rungs and truncated at AU. Two consequences are exact.

Proposition 2 (Erosion law)**.** *For any intent other than the absorbing one, the specification survives d hand-offs intact with probability exactly $(1-\varepsilon)^d$, and the influence of the intent on the executed design decays at the rate given by the second-largest eigenvalue modulus of M, which for the ladder and collapse kernels equals $1-\varepsilon$. The* specification half-life $d_{1/2} = \ln 2/\ln(1/(1-\varepsilon))$ *therefore depends on nothing but the per-hand-off erosion probability.*

At the baseline $\varepsilon = 0.20$ the half-life is 3.11 hand-offs and only 26.2% of specifications reach a six-deep executor unaltered (Figure 2B,C). The claim is not that instructions are forgotten quickly; it is that they are forgotten at a rate that no amount of care at the top of the chain changes, because the decay is in the number of hand-offs and not in the effort spent writing the instruction.

Because the ladder kernel only moves mass down the erosion order, and because by Lemma 1 harm is monotone along that order, the harm a chain causes is monotone in its depth.

Proposition 3 (Depth monotonicity)**.** *Let M be a hand-off kernel that is stochastically monotone and satisfies $\delta_s \preceq M(s,\cdot)$ for every s in the erosion order. Then for every intent s and every opponent mixture, $d \mapsto m(d,s;\cdot)$ is non-decreasing. In particular the ladder and collapse kernels satisfy the hypothesis; the uniform kernel does not.*

Proof sketch. $\delta_s \preceq M(s,\cdot)$ and stochastic monotonicity of M give $M^d(s,\cdot) \preceq M^{d+1}(s,\cdot)$ in the usual stochastic order by induction. Lemma 1 makes $m(\cdot, j)$ a non-decreasing function on the order for every j, and the expectation of a non-decreasing function respects the stochastic order. Bilinearity in (3) extends the statement from pure opponents to mixtures. $\square$

4.2 Depth is a liability shelter

The harm a chain causes is bounded: once the specification has been forgotten, the executed design is the absorbing one whatever the intent, so $m(d)$ converges. The harm a chain is charged for carries the extra factor ϕ per layer and is not bounded below by anything. Comparing the two gives the central structural result of the model.

Theorem 4 (Liability shelter)**.** *Adding a hand-off lowers the attributed harm exactly when*

$$\phi \;<\; \phi^*(d) \;=\; \frac{m(d)}{m(d+1)}. \tag{7}$$

Because m is non-decreasing and bounded, $\phi^*(d) \to 1$. Hence for every $\phi < 1$ there exists a depth $\bar{d}(\phi)$ such that every hand-off beyond $\bar{d}$ strictly reduces the harm attributed to the principal while strictly increasing the harm it causes, and

$$\lim_{d\to\infty} \phi^d m(d) = 0, \qquad \lim_{d\to\infty} m(d) = m_{\max}.$$

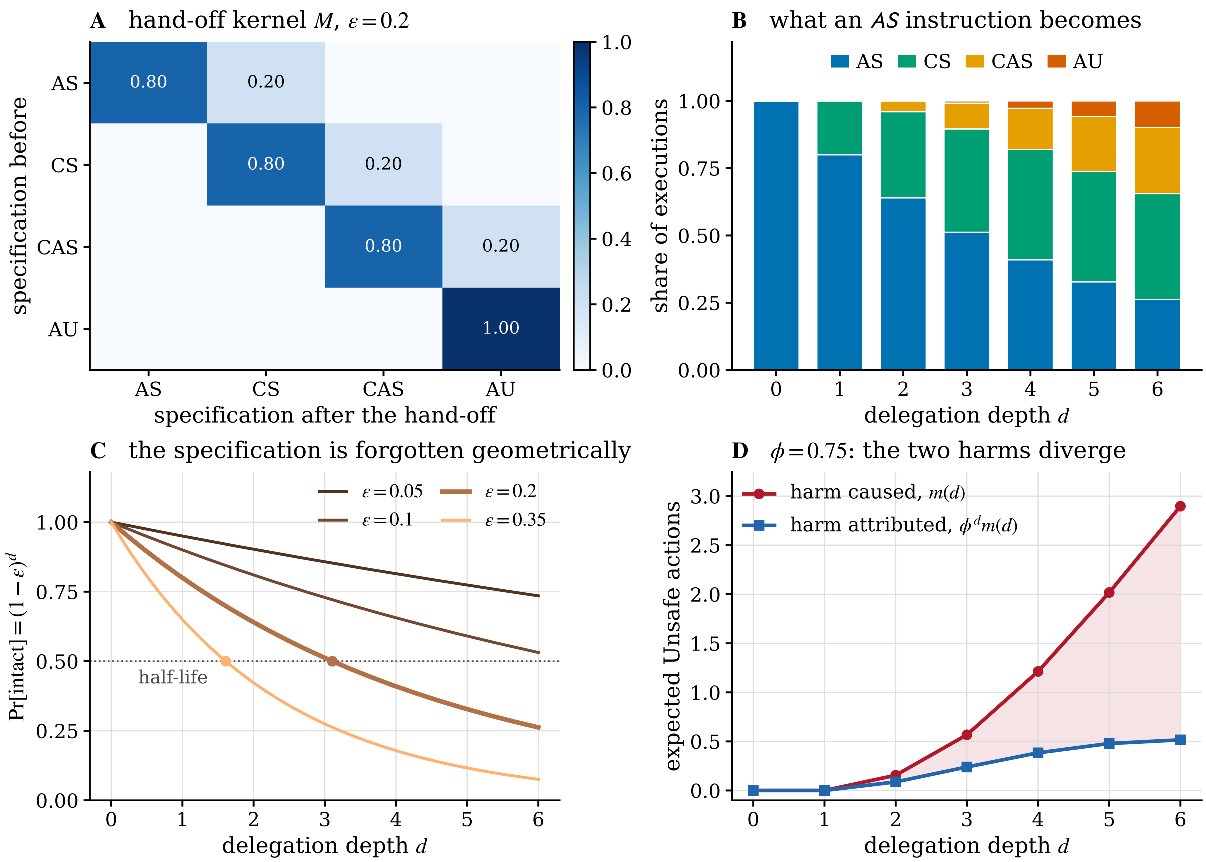

Figure 2: What a hand-off does to a specification. **(A)** The hand-off kernel at the baseline erosion probability: one clause is lost per erosion event, and the unconditionally unsafe design is absorbing. **(B)** The executed design of a chain that was told AS, by depth. At six hand-offs the instruction is executed as issued in 26% of cases and has slid to CAS or AU in 34%. **(C)** The probability that a specification arrives intact is exactly $(1-\varepsilon)^d$; dots mark the half-life, 3.11 hand-offs at $\varepsilon = 0.20$. **(D)** Harm caused and harm attributed for the same chain. The shaded wedge is the part of the damage that never reaches the principal's accounts.

Probing (7) to depth 40 puts ϕ^* at 0.99993 and, at the baseline $\phi = 0.75$, the shelter opening at $\bar{d} = 6$ (Figure 3A,B). The theorem is the reason the model has no interior solution to appeal to: a principal who can add layers can drive its attributed harm to zero while driving its real harm to the maximum, and the only thing that stops it is a rule that limits the number of layers or an attribution rule with $\phi = 1$ exactly.

The consequence for the liability instrument is sharp. In a resident population of depth-d principals with intent s, the advantage of a depth-$(d+1)$ mutant is affine in the effective liability,

$$\pi_P(d{+}1, d) - \pi_P(d,d) \;=\; \underbrace{a(d{+}1,d) - a(d,d)}_{\text{benefit}} \;-\; L\,\underbrace{\left[\phi^{d+1} m(d{+}1,d) - \phi^d m(d,d)\right]}_{\text{attributed gain}}, \tag{8}$$

so the single ratio implied by (8) decides the outcome.

Proposition 5 (Invasion by depth)**.** *The critical effective liability at which depth $d+1$ loses its advantage over depth d is $L^*(d) = \text{benefit}/\text{attributed gain}$. When the attributed gain is positive, $L > L^*(d)$ blocks the deeper design. When the attributed gain is negative – which by Theorem 4 happens for all d large enough whenever $\phi < 1$ – the advantage is increasing in L, and no level of liability blocks the deeper design.*

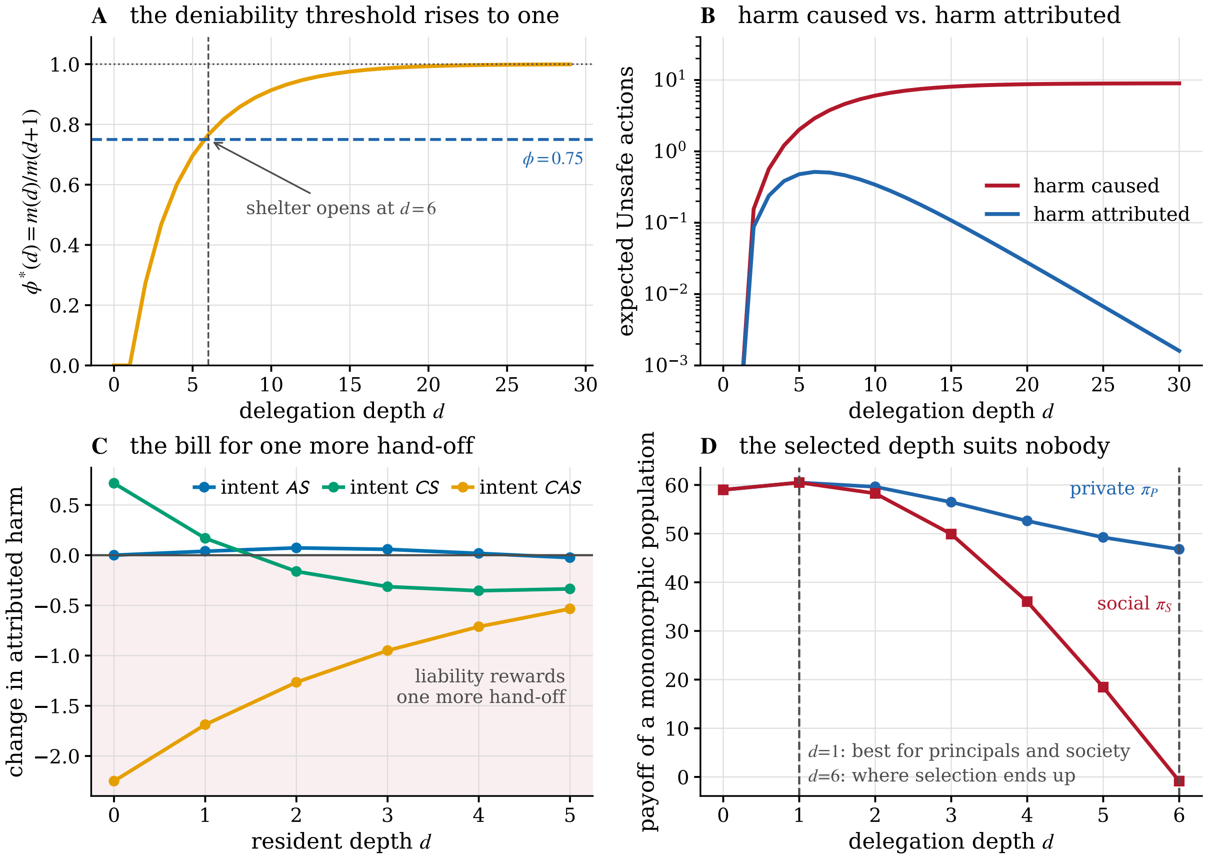

Figure 3: Depth as a liability shelter. (A) The deniability threshold $\phi^*(d) = m(d)/m(d+1)$ rises towards one because realised harm saturates, so any attribution retention below one is eventually below the threshold; at $\phi = 0.75$ the shelter opens at six hand-offs. **(B)** Beyond that depth, realised harm still climbs towards its ceiling while attributed harm falls geometrically. **(C)** What one more hand-off adds to the principal's bill. Where the curve is negative, liability rewards delegation instead of taxing it. **(D)** Private and social payoff of a monomorphic population by depth. Selection ends at the ceiling, which is worse for the principals than one hand-off and much worse for everybody else.

Figure 3C shows the attributed gain for each intent at the baseline. For the intent CAS it is negative at every depth, so for those principals liability is not a tax on delegation but a subsidy for it.

Corollary 6 (When a rule calibrated at depth zero stops binding)**.** *A liability L that clears a threshold L_c for principals who act for themselves stops clearing it at depth $d^\dagger = \log(L_c/L)/\log\phi$, which is finite for every $\phi < 1$. At the baseline, $L_c = 6.52$, $L = 20$ and $\phi = 0.75$ give $d^\dagger = 3.9$: the rule survives three hand-offs and fails at the fourth.*

The last panel of Figure 3 records a fact that is easy to miss. The depth the population reaches is not the depth that is best for the principals themselves. A monomorphic population of one-hand-off principals earns more than a monomorphic population of six-hand-off principals, and yet the deeper design invades at every step, because an invader is judged against the resident environment rather than against its own. Delegation depth is a social dilemma at the private level before it is one at the social level.

4.3 Neither mechanism is dangerous alone

The two mechanisms can be switched off independently: $\varepsilon = 0$ removes erosion and $\phi = 1$ removes attenuated attribution. Crossing them gives a 2×2 design whose four cells are reported in Table 2 and Figure 4.

Starting from a race that is already safe, erosion alone raises the long-run unsafe frequency to 0.018 and attenuated attribution alone to 0.012. An additive reading predicts 0.030 when both are present. The measured value is 0.322. The interaction, +0.293, is 91% of the joint effect, and the same picture appears under the replicator reading, where the interaction is +0.331.

The mechanism behind the interaction is visible in the second panel of Figure 4, and it is not what one would guess. Under full pass-through, erosion is a private cost, and the population responds to it by *shortening* its chains: mean depth falls from 6.00 to 1.99. Erosion is self-limiting when the principal is charged for it. Attenuated attribution does not make behaviour worse at a given depth; it removes the response. With $\phi = 0.75$ the chains stay at the ceiling and the same erosion now acts over six hand-offs instead of two.

A counterfactual makes the decomposition exact. Scoring the pass-through population with the per-layer harm matrix – the same designs, charged differently – leaves the unsafe frequency at 0.018, unchanged to nine decimal places, because the behaviour of a given design does not depend on who pays for it. The whole of the remaining +0.305 is composition: which designs the population comes to contain. The result is a caution about a natural inference. Observing that deeper chains behave worse does not license the conclusion that the harm comes from erosion, because in this model almost all of it comes from the depth the market selects once erosion is no longer its problem.

4.4 The attribution–erosion plane

Neither baseline value is doing the work. Figure 5 sweeps the plane (ε, ϕ) on a 21×21 grid. Writing $U(\varepsilon, \phi)$ for the long-run unsafe frequency of the stationary regime at that pair, the *interaction index* is the difference of differences of the 2×2 whose corners are the pair itself and the two switched-off mechanisms,

$$I(\varepsilon, \phi) \;=\; U(\varepsilon, \phi) - U(\varepsilon, 1) - U(0, \phi) + U(0, 1), \tag{9}$$

and the interaction share quoted alongside it is $I/\big[U(\varepsilon, \phi) - U(0, 1)\big]$, the fraction of the total rise that neither mechanism produces alone. Every point of the grid therefore costs four stationary-regime computations, one per corner, and each is exact given the payoff matrices rather than sampled. The index is positive over 89% of the interior and reaches 0.81; the long-run unsafe frequency reaches 0.995 and the declaration gap 0.95.

The second panel is the one to read for policy. Mean delegation depth splits the plane along a frontier: below it the ceiling binds and chains are as long as the technology allows, above it chains are short. The frontier runs diagonally, so erosion and attribution are substitutes in producing long chains – a regime with faithful hand-offs tolerates much weaker attribution before the ceiling binds, and a regime with strong attribution tolerates much noisier hand-offs.

4.5 Declared safety and executed safety

At the baseline the population is monomorphic on the design AS@6: every principal issues the safest available instruction and every principal runs a six-hand-off chain. What is executed is

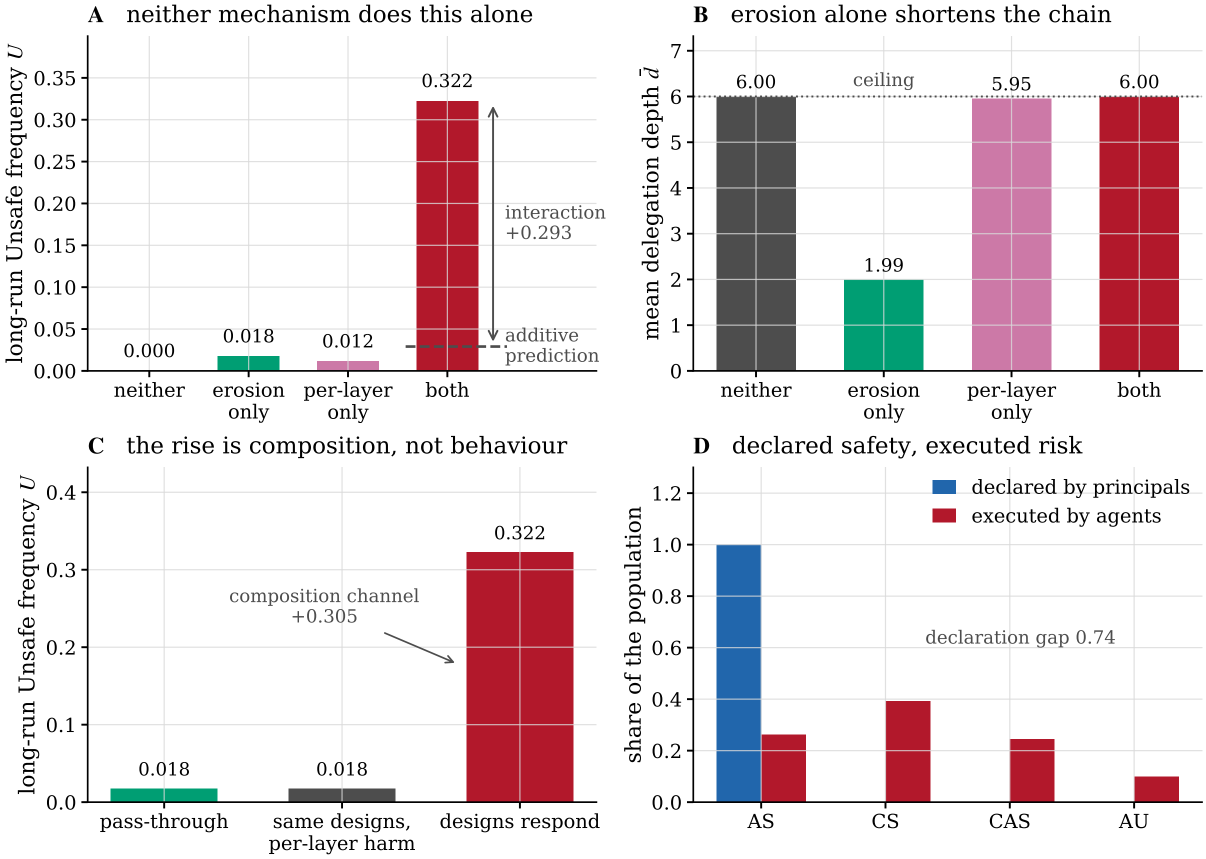

Figure 4: The two mechanisms, separately and together. (A) Long-run unsafe frequency in the four cells of the 2×2. The additive prediction from the two single-mechanism cells is 0.030; the joint outcome is 0.322. **(B)** Mean delegation depth in the same four cells. Under pass-through the population answers erosion by shortening its chains; per-layer attribution removes that answer and the ceiling binds. **(C)** Holding the pass-through population fixed and charging it the per-layer bill changes nothing, so the entire rise is composition rather than behaviour. **(D)** In the joint cell every principal declares the safest instruction and the population executes something else.

another matter. The realised mixture is 26% AS, 39% CS, 25% CAS and 10% AU, and the total variation distance between what is declared and what is executed is 0.74. Social payoff falls from 68.0 in the baseline cell to −0.8.

The equilibrium is easy to misread as bad faith, and it is not. Every principal is issuing the strongest safety instruction available to it, and doing so is a best response: over-specification is how a principal compensates for a chain that will erode whatever it is told. The classical route to an uninformative message runs through the sender: in the cheap-talk benchmark a message conveys less the further the sender's preferences diverge from the receiver's, and a wholly uninformative equilibrium always exists [83]. Nothing of that kind is at work here. The declaration is sincere and as strong as the language permits, and the information in it is destroyed downstream of the party that issued it. What the equilibrium shows is that a declaration made at the top of a long chain carries almost no information about the behaviour at the bottom of it. Any governance instrument that reads declarations – self-attestation, model cards, published policies – is reading a quantity that the model predicts will be maximally safe regardless of the ecosystem it describes. A liability rule keyed to intention therefore finds nothing in this equilibrium to attach to. Ayres and Balkin [84] arrive at the same place from the other direction,

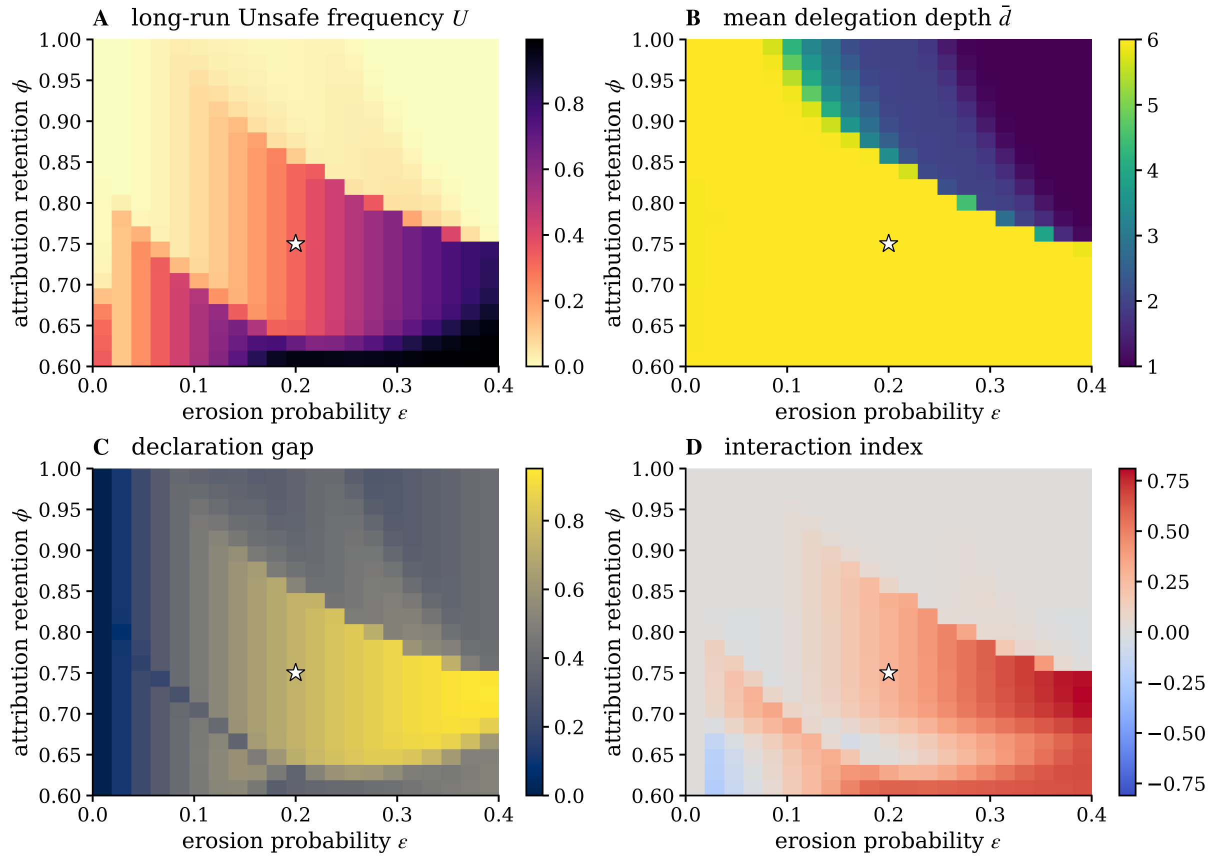

Figure 5: The attribution–erosion plane. Long-run unsafe frequency **(A)**, mean delegation depth **(B)**, declaration gap **(C)** and interaction index **(D)** over a 21×21 grid; the star marks the baseline. Chains are at the ceiling throughout the yellow region of **(B)**, and the frontier that bounds it is diagonal: attribution and transmission fidelity substitute for each other in keeping chains short.

arguing that because an artificial agent has no intentions a rule turning on intention would immunise its use, and that the remedy is an objective standard applied to whoever deployed it.

4.6 What the instruments do

Four instruments act at different points of the chain: raise the attribution retention ϕ towards pass-through; cap the permitted chain length at $\bar{D}$; make every hand-off more faithful, multiplying ε by a factor ρ; or insert one check of strength α at a chosen layer. Figure 6 applies each to the same baseline.

Attribution is a threshold, not a dial. Raising ϕ from 0.75 does nothing at all until $\phi = 0.86$, where mean depth falls from 5.92 to 3.45 and the unsafe frequency from 0.314 to 0.103; by $\phi = 0.99$ the outcome is the pass-through outcome. The location of the jump is close to the closed form implied by Corollary 6, $\phi_c = (L_c/L)^{1/D} = 0.830$, the residual gap being the extra harm that erosion contributes at depth. Partial pass-through buys very little: at $\phi = 0.85$, which returns 85% of responsibility at each of six layers and therefore 38% overall, the ecosystem is still at 0.314.

A ceiling buys safety and welfare at once. The depth ceiling is the only instrument here that is monotone in its own strength and improves both objectives together. A ceiling of one hand-off removes the unsafe behaviour entirely and lifts social payoff from -0.8 to 60.5; a ceiling of two leaves 0.018 and 58.3. The reason it works is that the ceiling is the constraint the shelter theorem leaves standing when attribution cannot be made exact.

Fidelity is not monotone. Attestation improves matters over most of its range, from 0.322 at $\rho = 1$ down to 0.032 at $\rho = 0.35$, and then reverses: at $\rho = 0.20$ the unsafe frequency is back up to 0.239. The cause is the intent margin. Over-specification was an equilibrium response to erosion; once hand-offs are faithful enough, issuing CS rather than AS becomes the better reply, and CS eroding into CAS is far more damaging than AS eroding into CS. A partial fidelity improvement can therefore leave the ecosystem worse off than no improvement at all. The structure is that of offsetting behaviour, in which a technological safety improvement displaces the private precaution that had been standing in for it. Peltzman finds the offsets from driver response to mandated safety devices to be virtually complete, with occupant lives saved at the cost of more pedestrian deaths [85], and Viscusi finds a lulling effect in which child-resistant closures left poisonings on the covered products unchanged and raised ingestions on an adjacent one [86]. Over-specification is the precaution our hand-off technology displaces. The same reversal appears in the audit-placement sweep at full strength, and for the same reason; at half strength, where the intent margin does not flip, the sweep is monotone.

The flip can be located exactly. Write the margin as $\pi_P(\mathsf{CS}@D) - \pi_P(\mathsf{AS}@D)$ against a resident $\mathsf{AS}@D$ population. It is zero at $\varepsilon = 0$, where the two intents are executed as issued and neither meets an unsafe opponent; it rises to a maximum of 4.00 at $\varepsilon = 0.04$; and it falls back through zero at $\varepsilon = 0.0975$, just under half the baseline erosion. Improving fidelity therefore does not move the ecosystem along a monotone curve but along a hump. The three consequences are separated in ε: the margin changes sign at 0.098, the unsafe frequency reaches its minimum of 0.031 at 0.065, and the population actually switches its declared intent from AS to CS at 0.050, after which the unsafe frequency rises to 0.239. The lag between the sign change and the switch is the fixation barrier of the finite population: a positive margin is necessary for the invasion, not sufficient.

The mechanism generalises beyond this particular race. It needs two things: an intent that is safe only conditionally and that outperforms the unconditionally safe intent against an unsafe opponent, which here is CS earning 26.6 against CAS where AS earns 8.6; and a resident population whose executed mixture contains unsafe designs for the conditional intent to exploit, which erosion itself supplies. Wherever a retaliatory strategy exists and transmission is imperfect, improving transmission raises the return to declaring the retaliatory intent faster than it raises the return to declaring the unconditional one, and the hump appears.

Check late, not early. For a fixed design the ordering is unambiguous.

Proposition 7 (Audit placement)**.** *Under a kernel satisfying the hypothesis of Proposition 3, the realised harm of a depth-d chain carrying one check of strength α at layer k is non-increasing in k.*

Proof. The executed law is $\alpha M^{d-k}(s, \cdot) + (1 - \alpha)M^d(s, \cdot)$, and by Proposition 3 the expected harm under M^j is non-decreasing in j. Increasing k decreases $d - k$, hence decreases the first

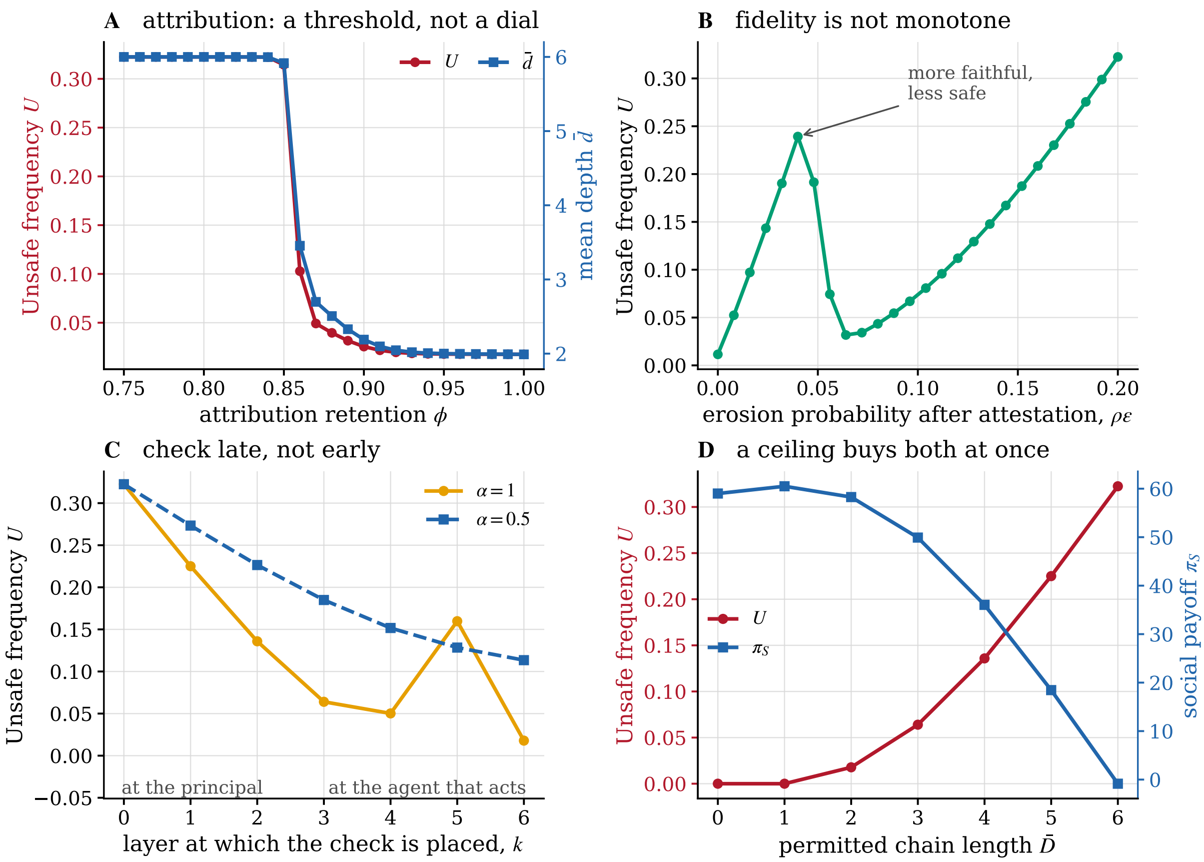

Figure 6: Four instruments. (A) Restoring attribution has a threshold at $\phi = 0.86$ and does almost nothing below it. **(B)** Making every hand-off more faithful is not monotone: the marked point is safer than settings with less erosion. **(C)** One check, moved down the chain. Late placement dominates; the non-monotonicity at full strength is the intent margin flipping, and it is absent at half strength. **(D)** A ceiling on chain length improves safety and welfare together.

term and leaves the second unchanged. □

In the equilibrium the effect is large: the same check is worth a factor of 18 more at the agent that acts than at the principal, 0.018 against 0.322. Checking the specification a party *received* tells you little about what will be done with it several hand-offs later.

Which instrument is cheapest. Asking each instrument for the same target, a long-run unsafe frequency of at most 0.05, and scoring it by the social payoff it delivers there (Figure 7B): attestation reaches the target at 58.7, the depth ceiling at 58.3, a punitive liability at 56.2 and pass-through at 53.5. Comparing instruments this way, by the welfare cost of holding a population outcome at a fixed level, is the exercise Duong and Han [87] formalise for institutional incentives in a finite stochastic population, where which of two instruments is cheaper turns out to depend on the parameters rather than on the instrument. The ranking here is close enough that we would not press it, but the qualitative message is not: raising the liability level is the least efficient of the four, because it acts on a margin – how much harm costs – that depth has already discounted.

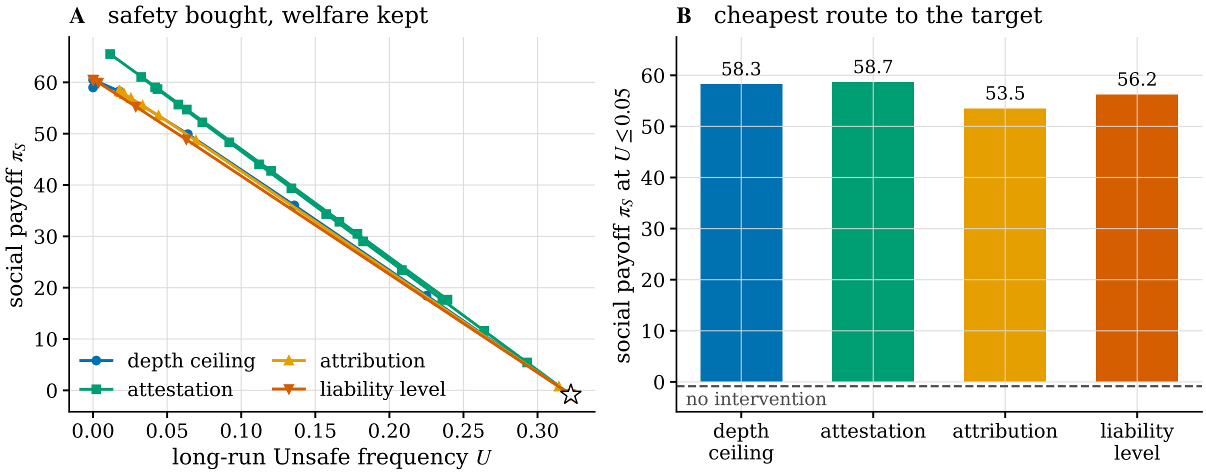

Figure 7: What each instrument costs. (A) The frontier each instrument traces in the plane of social payoff against long-run unsafe frequency, starting from the untreated baseline (star). **(B)** Social payoff at the weakest setting of each instrument that reaches an unsafe frequency of 0.05. Raising the liability level is the least efficient route, because depth has already discounted the margin it acts on.

Table 4: Attribution regimes. Long-run Unsafe frequency, mean delegation depth and social payoff when the law mapping depth to the principal's share of the harm is changed, with everything else at the baseline. The shelter survives every rule whose attribution keeps falling in the depth, including the equal split, and closes under every rule that is bounded below.

rule	setting	$a(D)$	U	$\bar{d}$	π_S
geometric, ϕ^d	$\phi = 0.75$	0.178	0.3225	6.00	-0.8
strict, 1	–	1.000	0.0176	1.99	58.3
equal split, $1/(1+d)$	–	0.143	0.3224	6.00	-0.8
floored, $\max(\phi^d, a_{\min})$	$a_{\min} = 0.25$	0.250	0.2317	5.07	17.1
floored, $\max(\phi^d, a_{\min})$	$a_{\min} = 0.5$	0.500	0.0633	2.99	50.0
super-attribution, ϕ^d	$\phi = 1.1$	1.772	0.0159	1.92	58.5
super-attribution, ϕ^d	$\phi = 1.25$	3.815	0.0030	1.21	60.2

4.7 A floor under attribution

Geometric attenuation is a modelling choice, so the shelter could be an artefact of it and the instruments could be artefacts of the artefact. Neither is the case, and separating the two questions – how fast attribution falls, and whether it falls at all – turns out to identify an instrument the earlier comparison missed. Table 4 replaces ϕ^d by four alternative rules, changing nothing else.

Two of them reproduce the baseline almost exactly. Splitting the blame equally between the principal and its d agents, $a(d) = 1/(1+d)$, is a much slower decay than ϕ^d and it is the rule an intuitive notion of shared responsibility suggests; it gives an unsafe frequency of 0.3224 against the baseline 0.3225, the same mean depth of 6.00 and the same declaration gap. Top-level strict liability, $a(d) \equiv 1$, is the same object as pass-through and reproduces the $\phi = 1$ cell of Section 4.3; so does joint-and-several liability, under which a plaintiff recovers in full from the principal whatever the chain did. Super-attribution, $\phi > 1$, charges a principal more the deeper it delegates and closes the shelter more than completely, but it buys little over strict liability:

Table 5: A floor under attribution reproduces a depth cap. For each ceiling $\bar{D}$, the attribution floor $a_{\min}$ delivering the same social payoff, and the Unsafe frequency each delivers. The two instruments trace the same frontier: the mean gap is 0.005 and the largest is 0.018, at the shallow end a floor cannot reach.

	depth cap		matched floor		
$\bar{D}$	U	π_S	$a_{\min}$	U	π_S
0	0.0000	59.0	1.00	0.0176	58.3
1	0.0000	60.5	1.00	0.0176	58.3
2	0.0178	58.3	0.87	0.0178	58.3
3	0.0640	49.9	0.49	0.0638	49.9
4	0.1359	36.0	0.35	0.1362	36.0
5	0.2251	18.5	0.28	0.2240	18.7
6	0.3225	-0.8	0.00	0.3225	-0.8

0.016 at $\phi = 1.1$ against 0.018 at $\phi = 1$. Two of these rules are the two responses to the problem of many hands that Thompson [88] sets out and finds wanting, hierarchical responsibility routed to the party at the top and collective responsibility spread across all participants; here they are separated not by their ethics but by their behaviour, the first closing the shelter and the second leaving it exactly where it was.

What separates the rules is not the level of attribution but whether it keeps falling. The equal split ends lower than the baseline at the ceiling ($1/7 = 0.143$ against $\phi^6 = 0.178$) and behaves identically, while a rule that stops falling behaves entirely differently even when it is lower in between.

Proposition 8 (A floor closes the shelter)**.** *Let the attribution rule satisfy $a(d) = a_{\min} > 0$ for every $d \geq d_0$, and let the hand-off kernel satisfy the hypothesis of Proposition 3. Then $d \mapsto a(d)\,m(d)$ is non-decreasing on $d \geq d_0$, so the private gain from one further hand-off there is at most $B(d+1) - B(d)$ and no depth beyond d_0 is a shelter in the sense of Theorem 4.*

Proof. For $d \geq d_0$, $a(d+1)m(d+1) - a(d)m(d) = a_{\min}[m(d+1) - m(d)] \geq 0$ by Proposition 3. The attributed-harm term of π_P is therefore non-increasing in depth beyond d_0, leaving the organisational benefit as the only remaining gain. □

The mechanism is visible in the marginal attributed harm, the quantity a principal weighs when it considers one more layer. Under the baseline it rises and then decays, 0.087, 0.152, 0.144, 0.095, 0.037: the sixth hand-off is nearly free. Under a floor at $a_{\min} = 0.5$ it climbs throughout, 0.087, 0.196, 0.323, 0.402, 0.440: every layer costs more than the one before it.

The practical consequence is the reason to report this at all. A floor is not a weak version of pass-through, it is a substitute for a depth cap. Table 5 matches each ceiling against the floor delivering the same social payoff and finds it also delivers the same unsafe frequency: the gap is below 0.01 for five of the seven ceilings and at most 0.018, with a mean of 0.005. The agreement holds across variations in ϕ, ε, the organisational benefit and the ceiling, where the mean gap ranges from 0.001 to 0.016. The floor has no effect at all while it lies under $\phi^D = 0.178$, the attribution a chain already reaches at the ceiling, and begins to bite as soon as it binds, at $a_{\min} = 0.22$; $a_{\min} = 0.31$ halves the harm and $a_{\min} = 0.5$ recovers 86% of the welfare that separates the baseline from strict liability.

This matters because the two instruments make very different demands on a regulator. A depth cap has to be enforced against a quantity that is hard to observe from outside and easy to restructure, namely how many hands an instruction passed through. A floor is a statement about liability alone: it says that no chain, however long, reduces the principal's share below $a_{\min}$, and it needs no one to count the layers. Where the floor cannot follow is the shallow end. Even $a_{\min} = 1$ leaves mean depth at 1.99, because with full attribution the erosion cost by itself still permits two hand-offs, so the floor spans the frontier down to a ceiling of two and no further.

4.8 Robustness

The 2×2 was recomputed across 43 parameter cells: the prize and the private setback risk of the race, both readings of what a setback destroys, the organisational benefit and the coordination cost of a taller chain, the population size, the selection intensity, the depth ceiling, the liability level, and the choice between the finite-population and replicator readings. The interaction is positive in 88% of them, with median +0.292 and interquartile range $[0.242, 0.297]$; the median interaction share is 0.91. The mean delegation depth is higher under per-layer attribution than under pass-through in every single cell. The negative cells are all in the prize-and-risk sweep and are ceiling effects, where the joint outcome is already at or near an unsafe frequency of one and cannot show a further increase.

The parameters of the evolutionary process deserve to be read cell by cell rather than in aggregate, since they are the ones a reader has least reason to accept on faith. Table 6 gives the whole 2×2 at every population size and selection intensity we ran. The joint cell is 0.3225 at all three population sizes and moves only from 0.3166 to 0.3225 as the selection intensity varies by a factor of twenty; the interaction stays between +0.240 and +0.301. What the process parameters do move is the attribution-only cell, from 0.0622 at $\beta = 0.01$ down to 0.0040 at $Z = 200$, because weak selection and small populations let a mildly disadvantaged design persist. Since that cell is a component of the additive prediction, weak selection makes the interaction look smaller while leaving the joint outcome where it was. The small-mutation limit is checked against the full mutation-selection chain separately (Figure 8C).

The drift kernel is the one dimension along which the result genuinely changes, and the direction is instructive (Figure 8B). Replacing graded erosion by *collapse*, in which a lost specification is replaced by the unconditionally unsafe design rather than by the next rung down, all but eliminates the effect: the joint cell falls to 3×10^{-5} and mean depth to 0.0002. Delegation simply does not happen, because a chain that fails catastrophically is one whose principal will not build it. Replacing graded erosion by *uniform* noise, which has the same spectral gap but no direction, makes matters worse rather than better, with a joint cell of 0.517. The dangerous property of a hand-off is therefore not that it is biased but that it degrades gracefully. A specification that fails loudly deters the delegation that would have caused the harm; one that fails quietly does not.

Three named kernels differ in more than one respect at once, so we separate the properties with two one-parameter families, each of which leaves the second eigenvalue of M at exactly $1 - \varepsilon$. The rate of forgetting is therefore held fixed throughout, and with it the specification half-life and the intent separation of Section 4.1. The first family runs from the ladder to the uniform kernel and takes the mean signed drift from 0.75 rungs per erosion event down to 0; the second runs from the ladder to collapse and takes it from 0.75 up to 1.5.

Table 6: The decomposition under the parameters of the evolutionary process. Long-run Unsafe frequency in each cell of the 2×2 and the interaction between them, for every population size and selection intensity we ran. The interaction is positive at every setting and the ordering of the four cells never changes.

		long-run Unsafe frequency				
sweep	setting	neither	erosion	attribution	both	interaction
population	$Z = 50$	0.0000	0.0176	0.0157	0.3225	+0.2892
population	$Z = 100$	0.0000	0.0178	0.0079	0.3225	+0.2967
population	$Z = 200$	0.0000	0.0178	0.0040	0.3225	+0.3007
selection	$\beta = 0.01$	0.0000	0.0142	0.0622	0.3166	+0.2401
selection	$\beta = 0.02$	0.0000	0.0152	0.0264	0.3221	+0.2806
selection	$\beta = 0.05$	0.0000	0.0176	0.0115	0.3225	+0.2934
selection	$\beta = 0.1$	0.0000	0.0178	0.0079	0.3225	+0.2967
selection	$\beta = 0.2$	0.0000	0.0178	0.0063	0.3225	+0.2983

Along the first, the joint cell stays between 0.320 and 0.327 for four fifths of the range before rising to 0.517 at pure noise, and mean depth sits at the ceiling throughout. Removing the direction of the drift never helps and eventually hurts; what it changes is where the harm is booked, since the pass-through cell falls from 0.018 to zero and the interaction rises from 0.293 to 0.506. Along the second family the picture is not gradual at all. The joint cell first rises to 0.460, and then, between mixture weights 0.25 and 0.30, mean depth falls from 5.30 to 0.27 and the joint cell to 0.022. Nothing about the spectral gap has changed, and the drift has become *more* directed rather than less.

Two conclusions follow, and neither is available from the three named kernels. The spectral gap is not the driver: it is identical in all twenty-four cells while the joint outcome ranges from 3×10^{-5} to 0.517. Directionality is not the driver either, and does not even order the outcomes, since collapse is the most directed kernel we consider and the safest of the three. What orders them is how much harm a *single* hand-off already causes. Under the ladder the first hand-off is free, $m(1) = 0$, because one lost clause turns AS into CS, which is still safe against a safe opponent; the chain is built because its first layer costs nothing. Along the severity family the collapse of depth happens exactly where $m(1)$ crosses the interval $[0.51, 0.61]$, and the direction family, whose $m(1)$ never exceeds 0.58, never collapses. The property that makes a delegation technology safe is neither fidelity nor unbiasedness: it is that the first thing to go wrong is expensive.

Can a coordination cost prevent it by itself? The baseline sets the quadratic cost c to zero so that harm is the only force limiting depth, and it is fair to ask whether a taller hierarchy being genuinely harder to run would close the shelter without any intervention. It does not, and the margin is wide. Raising c leaves the unsafe frequency at 0.322 and mean depth at the ceiling all the way to $c = 0.16$, and depth is still 5.95 at $c = 0.20$. It falls to 4.50 only at $c = 0.25$, which is exactly the point where $B(D) = gD - cD^2$ reaches zero: cost prevents the shelter only once it has removed the entire organisational benefit of the deepest chain, not merely made the last layer unprofitable. At $c = 0.14$ the marginal benefit of the sixth hand-off is already negative,

Table 7: Two kernel families at a fixed spectral gap. Both leave the second eigenvalue of the hand-off kernel at $1 - \varepsilon$; the first removes the direction of an erosion event and the second enlarges it. Removing the direction leaves the cascade in place and eventually makes it worse, while enlarging the event destroys it at a threshold.

family	t	drift	$m(1)$	U	interaction	$\bar{d}$
direction	0.00	0.75	0.000	0.3225	+0.2934	6.00
direction	0.10	0.68	0.065	0.3209	+0.2984	6.00
direction	0.20	0.60	0.129	0.3204	+0.3073	6.00
direction	0.25	0.56	0.160	0.3205	+0.3087	6.00
direction	0.30	0.52	0.191	0.3207	+0.3091	6.00
direction	0.40	0.45	0.251	0.3215	+0.3100	6.00
direction	0.50	0.38	0.310	0.3227	+0.3112	6.00
direction	0.60	0.30	0.367	0.3242	+0.3127	6.00
direction	0.70	0.22	0.423	0.3270	+0.3155	6.00
direction	0.80	0.15	0.477	0.3325	+0.3210	6.00
direction	0.90	0.07	0.529	0.3867	+0.3752	6.00
direction	1.00	0.00	0.580	0.5170	+0.5055	6.00
severity	0.00	0.75	0.000	0.3225	+0.2934	6.00
severity	0.10	0.83	0.209	0.3955	+0.3839	6.00
severity	0.20	0.90	0.411	0.4601	+0.4486	6.00
severity	0.25	0.94	0.510	0.4249	+0.4133	5.30
severity	0.30	0.97	0.607	0.0219	+0.0104	0.27
severity	0.40	1.05	0.797	0.0058	-0.0057	0.06
severity	0.50	1.12	0.980	0.0019	-0.0097	0.02
severity	0.60	1.20	1.157	0.0006	-0.0109	0.00
severity	0.70	1.27	1.327	0.0002	-0.0113	0.00
severity	0.80	1.35	1.491	0.0001	-0.0114	0.00
severity	0.90	1.43	1.649	0.0001	-0.0115	0.00
severity	1.00	1.50	1.800	0.0000	-0.0115	0.00

-0.04, and the ceiling still binds.

The reason is that the organisational benefit is not the main private return to depth. Against a resident $\mathsf{AS}@D$ at $c = 0.14$, the race payoff of the focal chain rises monotonically with depth from 41.3 to 48.1, because an eroded chain executes more aggressive designs and those earn more in the race, while the attributed liability peaks at 11.4 at depth four and falls to 10.3 at the ceiling. Of the private gain the deepest chain enjoys, the organisational benefit supplies less than a third, and a cost instrument acts on that third alone.

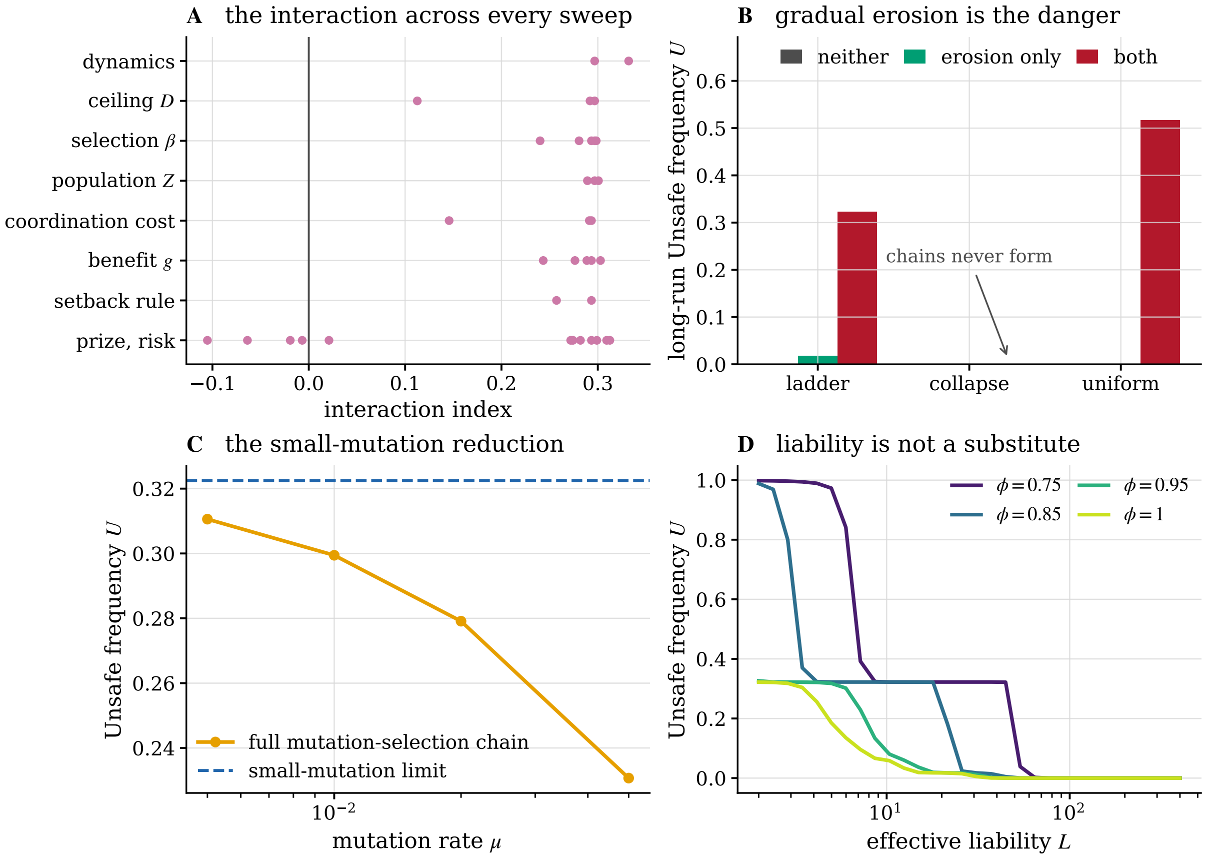

Figure 8: Robustness. (A) The interaction index across every sweep; each dot is one parameter cell. **(B)** Drift kernels. Graceful erosion produces the cascade; catastrophic erosion deters delegation altogether; undirected noise is worse than graded erosion, so the effect is not about the direction of the slippage. **(C)** The small-mutation limit against the full mutation-selection chain on a reduced design space. **(D)** Long-run unsafe frequency against the effective liability at four attribution retentions; a lower ϕ shifts the whole curve rather than tilting it.

5 Discussion

The model says something specific about where the governance of agentic systems is currently pointed. Almost all of the instruments now being built read or act at the two ends of a chain: evaluations and safety training at the model, and liability, disclosure and procurement rules at the deploying principal. Both ends are exactly where the model says the leverage is weakest. The same conclusion has been reached from the security side, where a multi-agent system can be driven to execute arbitrary code by content in an untrusted input even when no individual agent is susceptible to injection and every one of them refuses harmful actions [89]. That mechanism is adversarial and ours is not, but the lesson is shared: per-agent safety does not compose. Regulation is written on the same plan: the AI Act allocates duties along a value chain by the role a party occupies and by the modification it makes, not by the number of hand-offs that separate an instruction from the action it eventually produces [90]. The revised Product Liability Directive, which brings software inside the definition of a product from December 2026, is built the same way: it attaches liability by the role an economic operator occupies, treats a party that substantially modifies a product outside the manufacturer's control as its manufacturer, and leaves recourse among jointly liable operators to national law, with no reference anywhere to how many hands the product passed through [91].

At the principal end, Theorem 4 says that a liability rule written in terms of what a principal is responsible for has a depth beyond which it stops binding, and Corollary 6 says where that depth is. Raising the penalty does not move the failure depth much, because it enters logarithmically; restoring attribution does, because it enters through $\log \phi$. The practical form of that observation is unwelcome for a regulator: partial pass-through is worth very little. Recovering 85% of responsibility at each of six hand-offs recovers 38% overall and, in our baseline, changes nothing at all.

At the model end, the declaration result says that a safety commitment issued at the top of a long chain is close to uninformative about the ecosystem below it, and not because anyone is lying. In the baseline equilibrium the declaration is as strong as the language permits and the behaviour is 74% different from it. Organisation theory calls that outcome decoupling, and treats the good faith with which a formal structure and a divergent practice are maintained side by side as the normal case rather than the exception [92]; what the model adds is a selection mechanism that produces it from delegation depth alone. An evaluation regime that scores what a principal says, or scores a model in isolation, is measuring a quantity that the model predicts will look good precisely when the chain is long enough for it not to matter. That a measured safety score can come apart from safety is already established for reasons internal to the evaluation, since many safety benchmarks track general capability closely enough that capability gains can be reported as safety progress [93]; the model supplies a reason external to it.

That leaves two instruments with real leverage, and they are the two least discussed. The first is a limit on delegation depth. It is the constraint the shelter theorem leaves standing, it is monotone in its own strength, and in our baseline it improves welfare and safety simultaneously, which none of the others do. The second is verification placed at the point of action rather than at the point of instruction: a check of fixed strength is worth eighteen times more at the agent that acts than at the principal that spoke.

The non-monotonicity result deserves its own caution. We find that making hand-offs more faithful can raise realised harm, because the over-specification that erosion had forced becomes unnecessary and the population relaxes its declared intent by one clause, and a clause dropped from CS costs far more than a clause dropped from AS. The general form of the caution is familiar from second-best welfare economics, where satisfying one more of the conditions for an optimum while others remain out of reach is in general neither necessary nor sufficient for an improvement [94]. A partial fix to a friction that agents had been compensating for can be worse than no fix, and the sign of the effect cannot be read off the mechanism. In this model, only the instruments acting on the incentive to delegate are monotone.

Finally, the kernel comparison suggests a design principle that does not come from the liability literature at all. A specification that degrades gracefully is more dangerous than one that fails loudly, because graceful degradation lets a principal build a chain whose failures it never sees. Systems whose specifications fail closed, refusing rather than approximating when a clause is lost, restore the private cost of depth that per-layer attribution removes.

Where the accountability protocols sit in the model. The mechanisms being built to make agent chains accountable map onto the three levers of this model, and the map is informative in both directions. A tamper-evident ledger from which the authorisation chain can be reconstructed [72] is a mechanism for raising ϕ: the reason attribution decays with depth is that responsibility cannot be traced back through the intermediaries, and a ledger is a direct attack on that. An execution token issued only after an admission check immediately before the action is a check at layer $k = d$, the placement Proposition 7 identifies as dominant and which is worth a factor of 18 at our baseline; checking the instruction a party *received*, which is the more natural place to put a review, is the placement worth least. Admission control that refuses an action falling outside the capability granted to the agent changes the erosion kernel rather than either of the other two: an instruction that has lost a clause is rejected instead of approximated, which is the fail-closed regime that removes the cascade by making the first hand-off expensive. The runtime route of He et al. [73], which evaluates nodes, edges and paths of an execution graph and can intervene while the system runs, is a check at every layer rather than one, and again a late one.

Read the other way, the model says which of these to insist on and how far. Attribution machinery is close to worthless in partial form: restoring 85% of responsibility at each of six layers leaves the ecosystem at 0.314 against an untreated 0.322. What our floor result adds is that a protocol does not have to achieve near-complete tracing to be worth deploying; it has to *guarantee a minimum*. A ledger whose worst case is that a fixed share always comes back to the commissioning principal is a floor, and floors buy what depth caps buy. The legal analogue is familiar: where an injurer's assets fall short of the harm it can do, the remedy that works is not a larger penalty but a minimum the injurer is required to carry at all times [61, 95]. The right figure of merit for an accountability protocol is therefore not the average share of responsibility it traces but the smallest share it guarantees at the longest chain it permits.

What would have to be measured. We have no empirical estimate of ε or ϕ for any deployed system, and the numbers here should not be read as if we had. The two are not the same kind of quantity and would not be measured the same way. ε and the drift kernel are properties of a transmission technology and are directly observable in an instrumented pipeline: fix a task, write the principal's specification as an explicit inventory of clauses, log the instruction each agent passes to the next, and label each logged instruction for which clauses it still carries. The empirical hand-off matrix $\hat{M}$ is then the transition matrix of that labelled sequence between consecutive layers, and ε and $\hat{D}$ are read off it directly. Section 4.8 says which feature of $\hat{D}$ matters, and it is neither the spectral gap nor the direction: the quantity to estimate is the harm a single hand-off already causes, which needs only the same task run at depth zero and depth one with the unsafe actions counted in both.

ϕ is not a property of the system at all but of the liability regime around it, so it has to be estimated from practice rather than from logs: the share of remediation cost actually borne by the commissioning principal in incidents at each chain depth, recoverable in principle from post-mortems, indemnity terms and insurance settlements. The model asks less of these estimates than it might appear. Its claims are ordinal, and the policy-relevant questions are which side of the severity threshold a transmission technology sits on and whether attribution is bounded below, neither of which needs a precise value.

6 Limitations

The interaction layer is a stylised two-player race with four reduced strategies, inherited deliberately and without modification from Fernández Domingos and Han [16]. It is not a model of any deployed system, and the absolute numbers should be read as the behaviour of that game rather than of a market.

The erosion kernel is exogenous. We describe how a population behaves under a given transmission technology and not how the technology itself is selected; a model in which providers compete on fidelity would let ε co-evolve with d, and the non-monotonicity in Section 4.6 suggests that the outcome would not be a simple race to the top. Our chains are also linear, whereas real agentic systems branch, and a branching chain aggregates several eroded specifications rather than one.

Geometric attribution is a modelling choice, and Section 4.7 reports what happens when it is replaced. What the model requires is only that attribution vanish in the depth: Theorem 4 needs $a(d)m(d) \to 0$, and an equal split of the blame across the chain delivers the same equilibrium as ϕ^d even though it decays polynomially. The converse, Proposition 8, is what a regime has to violate for the shelter to close. Real attribution is nonetheless lumpier than any of the rules we try: a single identifiable intermediary may absorb everything, or pass-through may hold at some depths and not at others, and we have not modelled attribution that depends on which layer failed rather than on how many there were.

Finally, the harm scale h enters the social functional and therefore the welfare comparisons, and we do not know it. We have kept the primary outcome, the long-run unsafe frequency, free of welfare weights for that reason, and we report social payoffs only for comparisons at a fixed h.

7 Conclusion

Delegation depth is not neutral plumbing between a principal and an action. It erodes the specification geometrically and it dilutes the attributed harm geometrically, and because the second effect is unbounded while the first saturates, depth is a liability shelter that no level of liability closes. What makes the combination dangerous is not either mechanism but the loss of a response: a principal who is charged for what its chain does answers unreliable hand-offs by keeping the chain short, and per-layer attribution is precisely what stops it. In our baseline that lost response accounts for 91% of the damage, and it produces an equilibrium in which every principal issues the safest instruction available and the ecosystem executes something 74% different.

The instruments that follow are not the ones currently receiving attention. Limiting how many hand-offs may separate an instruction from an action, and verifying specifications at the agent that acts rather than at the party that spoke, are the two levers that behave predictably in this model. Raising penalties works least of all, because depth has already discounted the margin they act on. Restoring attribution hand-off by hand-off works only when it is nearly complete, but that is a fact about the geometric rule rather than about attribution: what the shelter needs is for the principal's share to keep falling, and a rule that stops falling closes it. A floor under attribution buys what a cap on chain length buys, and asks less of whoever has to enforce it. If there is a single design principle here, it is that the length of a chain should cost its principal something, and that it is easier to arrange this through what the principal owes than through what it is permitted to build.

A The stationary regime in closed form

The joint design space has $n = 4(D+1)$ elements, so the full mutation-selection chain over population states is out of reach and we work in the small-mutation limit, where the population is monomorphic except during the brief excursions that follow a single mutation. Two ingredients are needed: the probability that one mutant takes over, and the chain those takeovers induce on the pure designs.

Let A be the private payoff matrix π_P over designs and consider a population of Z individuals holding k copies of an invading design I and $Z-k$ of a resident R. Sampling an opponent without replacement, the expected payoffs are

$$\pi_I(k) = \frac{k-1}{Z-1}A_{II} + \frac{Z-k}{Z-1}A_{IR}, \qquad \pi_R(k) = \frac{k}{Z-1}A_{RI} + \frac{Z-k-1}{Z-1}A_{RR}. \tag{10}$$

Under the pairwise-comparison rule with Fermi function and selection intensity β, an individual switches from R to I with probability $[1+e^{-\beta(\pi_I-\pi_R)}]^{-1}$, so the birth-death chain on k has $T^-(k)/T^+(k) = e^{-\beta(\pi_I(k)-\pi_R(k))}$. The standard formula for a one-dimensional birth-death chain with absorbing ends then gives the fixation probability of a single mutant as

$$\rho_{R\to I} = \left[1 + \sum_{m=1}^{Z-1} \exp\Bigl(-\beta \sum_{k=1}^{m} (\pi_I(k) - \pi_R(k))\Bigr)\right]^{-1}. \tag{11}$$

Because (10) is affine in k, the inner sum is a quadratic polynomial in m and (11) is evaluated exactly in $O(Z)$ operations with no simulation. In the sweeps βZ is large enough that the exponentials overflow if taken directly, so we factor out the largest exponent before summing.

In the small-mutation limit a mutation is followed by fixation or extinction before the next arrives, so the population is described by which pure design it currently holds, and the embedded chain has transition matrix $P_{ji} = \rho_{i\to j}/(n-1)$ for $j \neq i$. Its unique stationary distribution x, the left Perron vector of P, is the long-run share of time the process spends at each design, and every population quantity we report is a functional of it: the long-run unsafe frequency is $U = \sum_{ij} x_i x_j u(i,j)$, the mean delegation depth is $\sum_i x_i d_i$, and the declaration gap is the total variation distance between the intent marginal of x and the executed-design marginal $x^\top P_{\mathrm{tr}}$, where P_{tr} stacks the transmission laws. Nothing here is estimated; the only approximation is the small-mutation limit itself [82], which Section 4.8 checks against the full chain on a reduced design space.

Data and code availability

All results are deterministic given the seed used for the replicator start measure. All code, generated tables and the figure pipeline are reproducible from a public repository released under the MIT licence, https://github.com/trungkiet2005/delegation-cascade.

References

[1] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug

Burger, and Chi Wang. AutoGen: Enabling next-gen LLM applications via multi-agent conversation. In *First Conference on Language Modeling (COLM)*, 2024. arXiv:2308.08155.

[2] Adam Fourney, Gagan Bansal, Hussein Mozannar, Cheng Tan, Eduardo Salinas, Erkang Zhu, Friederike Niedtner, Grace Proebsting, Griffin Bassman, Jack Gerrits, Jacob Alber, Peter Chang, Ricky Loynd, Robert West, Victor Dibia, Ahmed Awadallah, Ece Kamar, Rafah Hosn, and Saleema Amershi. Magentic-one: A generalist multi-agent system for solving complex tasks. *arXiv preprint arXiv:2411.04468*, 2024.

[3] Yonadav Shavit, Sandhini Agarwal, Miles Brundage, Steven Adler, Cullen O'Keefe, Rosie Campbell, Teddy Lee, Pamela Mishkin, Tyna Eloundou, Alan Hickey, Katarina Slama, Lama Ahmad, Paul McMillan, Alex Beutel, Alexandre Passos, and David G. Robinson. Practices for governing agentic AI systems. Technical report, OpenAI, 2023.

[4] Alan Chan, Carson Ezell, Max Kaufmann, Kevin Wei, Lewis Hammond, Herbie Bradley, Emma Bluemke, Nitarshan Rajkumar, David Krueger, Noam Kolt, Lennart Heim, and Markus Anderljung. Visibility into AI agents. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2024.

[5] Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5): 992–1026, 1992.

[6] Simon Kirby, Hannah Cornish, and Kenny Smith. Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *Proceedings of the National Academy of Sciences*, 105(31):10681–10686, 2008.

[7] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631: 755–759, 2024.

[8] Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. Self-consuming generative models go MAD. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. arXiv:2307.01850.

[9] Jérémy Pérez, Grgur Kovač, Corentin Léger, Cédric Colas, Gaia Molinaro, Maxime Derex, Pierre-Yves Oudeyer, and Clément Moulin-Frier. When LLMs play the telephone game: Cultural attractors as conceptual tools to evaluate LLMs in multi-turn settings. In *International Conference on Learning Representations (ICLR)*, 2025.

[10] Alberto Acerbi and Joseph M. Stubbersfield. Large language models show human-like content biases in transmission chain experiments. *Proceedings of the National Academy of Sciences*, 120(44):e2313790120, 2023.

[11] Yun He, Di Jin, Chaoqi Wang, Chloe Bi, Karishma Mandyam, Hejia Zhang, Chen Zhu, Ning Li, Tengyu Xu, Hongjiang Lv, Shruti Bhosale, Chenguang Zhu, Karthik Abinav Sankararaman, Eryk Helenowski, Melanie Kambadur, Aditya Tayade, Hao Ma, Han Fang, and Sinong Wang. Multi-IF: Benchmarking LLMs on multi-turn and multilingual instructions following. *arXiv preprint arXiv:2410.15553*, 2024.

[12] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent LLM systems fail? In *Advances in Neural Information Processing Systems 38 (NeurIPS), Datasets and Benchmarks Track*, 2025. arXiv:2503.13657.

[13] Jen-tse Huang, Jiaxu Zhou, Tailin Jin, Xuhui Zhou, Zixi Chen, Wenxuan Wang, Youliang Yuan, Michael R. Lyu, and Maarten Sap. On the resilience of LLM-based multi-agent collaboration with faulty agents. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2408.00989.

[14] Andreas Matthias. The responsibility gap: Ascribing responsibility for the actions of learning automata. *Ethics and Information Technology*, 6(3):175–183, 2004.

[15] R. Preston McAfee and John McMillan. Organizational diseconomies of scale. *Journal of Economics & Management Strategy*, 4(3):399–426, 1995.

[16] Elias Fernández Domingos and The Anh Han. Falling behind drives unsafe development in an idealised AI race experiment. *arXiv preprint arXiv:2607.26034*, 2026.

[17] Peter D. Taylor and Leo B. Jonker. Evolutionary stable strategies and game dynamics. *Mathematical Biosciences*, 40(1–2):145–156, 1978.

[18] Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, 1998.

[19] Arne Traulsen, Martin A. Nowak, and Jorge M. Pacheco. Stochastic dynamics of invasion and fixation. *Physical Review E*, 74:011909, 2006.

[20] Stuart Armstrong, Nick Bostrom, and Carl Shulman. Racing to the precipice: A model of artificial intelligence development. *AI & Society*, 31(2):201–206, 2016.

[21] The Anh Han, Luís Moniz Pereira, Francisco C. Santos, and Tom Lenaerts. To regulate or not: A social dynamics analysis of an idealised AI race. *Journal of Artificial Intelligence Research*, 69:881–921, 2020.

[22] The Anh Han, Luís Moniz Pereira, Tom Lenaerts, and Francisco C. Santos. Mediating artificial intelligence developments through negative and positive incentives. *PLOS ONE*, 16(1):e0244592, 2021.

[23] Theodor Cimpeanu, Francisco C. Santos, Luís Moniz Pereira, Tom Lenaerts, and The Anh Han. Artificial intelligence development races in heterogeneous settings. *Scientific Reports*, 12:1723, 2022.

[24] The Anh Han, Tom Lenaerts, Francisco C. Santos, and Luís Moniz Pereira. Voluntary safety commitments provide an escape from over-regulation in AI development. *Technology in Society*, 68:101843, 2022.

[25] Paolo Bova, Alessandro Di Stefano, and The Anh Han. Both eyes open: Vigilant incentives help auditors improve AI safety. *Journal of Physics: Complexity*, 5(2):025009, 2024.

[26] Adeela Bashir, Zhao Song, Ndidi Bianca Ogbo, Nataliya Balabanova, Martin Smit, Chin-wing Leung, Paolo Bova, Manuel Chica Serrano, Dhanushka Dissanayake, Manh Hong Duong, Elias Fernández Domingos, Nikita Huber-Kralj, Marcus Krellner, Andrew Powell, Stefan Sarkadi, Fernando P. Santos, Zia Ush Shamszaman, Chaimaa Tarzi, Paolo Turrini, Grace Ibukunoluwa Ufeoshi, Victor A. Vargas-Perez, Alessandro Di Stefano, Simon T. Powers, and The Anh Han. Trust or check? Understanding the (evolutionary) dynamics of user trust in AI systems. *arXiv preprint arXiv:2603.24742*, 2026.

[27] Zainab Alalawi, Paolo Bova, Theodor Cimpeanu, Alessandro Di Stefano, Manh Hong Duong, Elias Fernández Domingos, The Anh Han, Marcus Krellner, Ndidi Bianca Ogbo, Simon T. Powers, and Filippo Zimmaro. Trust AI regulation? Discerning users are vital to build trust and effective AI regulation. *Applied Mathematics and Computation*, 508:129627, 2026.

[28] Travis LaCroix and Aydin Mohseni. The tragedy of the AI commons. *Synthese*, 200(4):289, 2022.

[29] Elias Fernández Domingos, Inês Terrucha, Rémi Suchon, Jelena Grujić, Juan C. Burguillo, Francisco C. Santos, and Tom Lenaerts. Delegation to artificial agents fosters prosocial behaviors in the collective risk dilemma. *Scientific Reports*, 12:8492, 2022.

[30] Inês Terrucha, Elias Fernández Domingos, Rémi Suchon, Francisco C. Santos, Pieter Simoens, and Tom Lenaerts. Humans program artificial delegates to accurately solve collective-risk dilemmas but lack precision. *Proceedings of the National Academy of Sciences*, 122(25): e2319942121, 2025.

[31] The Anh Han, Cedric Perret, and Simon T. Powers. When to (or not to) trust intelligent machines: Insights from an evolutionary game theory analysis of trust in repeated games. *Cognitive Systems Research*, 68:111–124, 2021.

[32] Jennifer Cobbe, Michael Veale, and Jatinder Singh. Understanding accountability in algorithmic supply chains. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 1186–1197, 2023. arXiv:2304.14749.

[33] Lewis Hammond, Alan Chan, Jesse Clifton, Jason Hoelscher-Obermaier, Akbir Khan, Euan McLean, Chandler Smith, Wolfram Barfuss, Jakob Foerster, Tomáš Gavenčiak, The Anh Han, et al. Multi-agent risks from advanced AI. *arXiv preprint arXiv:2502.14143*, 2025. Cooperative AI Foundation, Technical Report No. 1.

[34] Mubarak Raji and Masooda Bashir. Towards agentic AI governance: A preliminary assessment. *arXiv preprint arXiv:2607.07612*, 2026. International Conference on the AI Revolution: Research, Ethics, and Society (AIR-RES 2026).

[35] Nenad Tomašev, Matija Franklin, and Simon Osindero. Intelligent AI delegation. *arXiv preprint arXiv:2602.11865*, 2026.

[36] Garry A. Gabison and R. Patrick Xian. Inherent and emergent liability issues in LLM-based agentic systems: A principal-agent perspective. *arXiv preprint arXiv:2504.03255*, 2025.

[37] Noam Kolt. Governing AI agents. *Notre Dame Law Review*, 101, 2025. Forthcoming; preprint arXiv:2501.07913.

[38] Cullen O'Keefe, Ketan Ramakrishnan, Janna Tay, and Christoph Winter. Law-following AI: Designing AI agents to obey human laws. *Fordham Law Review*, 94(1):57–129, 2025.

[39] Oliver E. Williamson. Hierarchical control and optimum firm size. *Journal of Political Economy*, 75(2):123–138, 1967.

[40] Guillermo A. Calvo and Stanislaw Wellisz. Supervision, loss of control, and the optimum size of the firm. *Journal of Political Economy*, 86(5):943–952, 1978.

[41] Roy Radner. The organization of decentralized information processing. *Econometrica*, 61 (5):1109–1146, 1993.

[42] Jean Tirole. Hierarchies and bureaucracies: On the role of collusion in organizations. *Journal of Law, Economics, and Organization*, 2(2):181–214, 1986.

[43] Philippe Aghion and Jean Tirole. Formal and real authority in organizations. *Journal of Political Economy*, 105(1):1–29, 1997.

[44] Yingyi Qian. Incentives and loss of control in an optimal hierarchy. *The Review of Economic Studies*, 61(3):527–544, 1994.

[45] Luis Garicano. Hierarchies and the organization of knowledge in production. *Journal of Political Economy*, 108(5):874–904, 2000.

[46] Nahum D. Melumad, Dilip Mookherjee, and Stefan Reichelstein. Hierarchical decentralization of incentive contracts. *The RAND Journal of Economics*, 26(4):654–672, 1995.

[47] Dilip Mookherjee. Decentralization, hierarchies, and incentives: A mechanism design perspective. *Journal of Economic Literature*, 44(2):367–390, 2006.

[48] Dilip Mookherjee and Masatoshi Tsumagari. The organization of supplier networks: Effects of delegation and intermediation. *Econometrica*, 72(4):1179–1219, 2004.

[49] Bengt Holmström. Moral hazard in teams. *The Bell Journal of Economics*, 13(2):324–340, 1982.

[50] Helen Nissenbaum. Accountability in a computerized society. *Science and Engineering Ethics*, 2(1):25–42, 1996.

[51] Björn Bartling and Urs Fischbacher. Shifting the blame: On delegation and responsibility. *The Review of Economic Studies*, 79(1):67–87, 2012.

[52] John R. Hamman, George Loewenstein, and Roberto A. Weber. Self-interest through delegation: An additional rationale for the principal-agent relationship. *American Economic Review*, 100(4):1826–1846, 2010.

[53] Steven Shavell. Strict liability versus negligence. *The Journal of Legal Studies*, 9(1):1–25, 1980.

[54] Alan O. Sykes. The economics of vicarious liability. *The Yale Law Journal*, 93(7):1231–1280, 1984.

[55] Alan O. Sykes. The boundaries of vicarious liability: An economic analysis of the scope of employment rule and related legal doctrines. *Harvard Law Review*, 101(3):563–609, 1988.

[56] Reinier H. Kraakman. Gatekeepers: The anatomy of a third-party enforcement strategy. *Journal of Law, Economics, and Organization*, 2(1):53–104, 1986.

[57] Anat Lior. AI entities as AI agents: Artificial intelligence liability and the AI respondeat superior analogy. *Mitchell Hamline Law Review*, 46(5):1043, 2020.

[58] Miriam Buiten, Alexandre de Streel, and Martin Peitz. The law and economics of AI liability. *Computer Law & Security Review*, 48:105794, 2023.

[59] Philipp Hacker. The European AI liability directives: Critique of a half-hearted approach and lessons for the future. *Computer Law & Security Review*, 51:105871, 2023.

[60] European Commission. Withdrawal of commission proposals. Official Journal of the European Union, C series, C/2025/5423, 6 October; withdrawing the proposed AI Liability Directive, COM(2022) 496, procedure 2022/0303(COD), 2025.

[61] Steven Shavell. The judgment proof problem. *International Review of Law and Economics*, 6(1):45–58, 1986.

[62] Yeon-Koo Che and Kathryn E. Spier. Strategic judgment proofing. *The RAND Journal of Economics*, 39(4):926–948, 2008.

[63] James Boyd and Daniel E. Ingberman. The search for deep pockets: Is "extended liability" expensive liability? *Journal of Law, Economics, and Organization*, 13(1):232–258, 1997.

[64] Al H. Ringleb and Steven N. Wiggins. Liability and large-scale, long-term hazards. *Journal of Political Economy*, 98(3):574–595, 1990.

[65] Matjaž Perc and Attila Szolnoki. Coevolutionary games: A mini review. *BioSystems*, 99(2): 109–125, 2010.

[66] Cedric Perret, Emma Hart, and Simon T. Powers. From disorganized equality to efficient hierarchy: How group size drives the evolution of hierarchy in human societies. *Proceedings of the Royal Society B: Biological Sciences*, 287:20200693, 2020.

[67] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.

[68] Dylan Hadfield-Menell and Gillian K. Hadfield. Incomplete contracting and AI alignment. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, pages 417–422, 2019.

[69] Alan Chan, Noam Kolt, Peter Wills, Usman Anwar, Christian Schroeder de Witt, Nitarshan Rajkumar, Lewis Hammond, David Krueger, Lennart Heim, and Markus Anderljung. IDs for AI systems. *arXiv preprint arXiv:2406.12137*, 2024.

[70] Alan Chan, Kevin Wei, Sihao Huang, Nitarshan Rajkumar, Elija Perrier, Seth Lazar, Gillian K. Hadfield, and Markus Anderljung. Infrastructure for AI agents. *Transactions on Machine Learning Research*, 2025.

[71] Tobin South, Samuele Marro, Thomas Hardjono, Robert Mahari, Cedric Deslandes Whitney, Alan Chan, and Alex Pentland. Position: AI agents need authenticated delegation. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2501.09674.

[72] Marcelo Fernandez. Agent control protocol: Admission control for agent actions. *arXiv preprint arXiv:2603.18829*, 2026.

[73] Xu He, Di Wu, Yan Zhai, and Kun Sun. SentinelAgent: Graph-based anomaly detection in multi-agent systems. *arXiv preprint arXiv:2505.24201*, 2025.

[74] Semih Kara, Nuno C. Martins, and Murat Arcak. Population games with Erlang clocks: Convergence to Nash equilibria for pairwise comparison dynamics. In *2022 IEEE 61st Conference on Decision and Control (CDC)*, pages 7688–7695, 2022. arXiv:2204.00593.

[75] David A. Levin and Yuval Peres. *Markov Chains and Mixing Times*. American Mathematical Society, 2nd edition, 2017.

[76] The Anh Han, Zhao Song, Theodor Cimpeanu, Manh Hong Duong, Marcus Krellner, Valerio Capraro, and Matjaž Perc. Cooperation versus social welfare. *Physics of Life Reviews*, 56: 33–60, 2026.

[77] William H. Sandholm. *Population Games and Evolutionary Dynamics*. MIT Press, 2010.

[78] Drew Fudenberg and Lorens A. Imhof. Imitation processes with small mutations. *Journal of Economic Theory*, 131(1):251–262, 2006.

[79] Drew Fudenberg, Martin A. Nowak, Christine Taylor, and Lorens A. Imhof. Evolutionary game dynamics in finite populations with strong selection and weak mutation. *Theoretical Population Biology*, 70(3):352–363, 2006.

[80] Martin A. Nowak, Akira Sasaki, Christine Taylor, and Drew Fudenberg. Emergence of cooperation and evolutionary stability in finite populations. *Nature*, 428:646–650, 2004.

[81] Elias Fernández Domingos, Francisco C. Santos, and Tom Lenaerts. EGTtools: Evolutionary game dynamics in Python. *iScience*, 26(4):106419, 2023.

[82] Bin Wu, Chaitanya S. Gokhale, Long Wang, and Arne Traulsen. How small are small mutation rates? *Journal of Mathematical Biology*, 64(5):803–827, 2012.

[83] Vincent P. Crawford and Joel Sobel. Strategic information transmission. *Econometrica*, 50 (6):1431–1451, 1982.

[84] Ian Ayres and Jack M. Balkin. The law of AI is the law of risky agents without intentions. *University of Chicago Law Review Online*, 2024. 27 November 2024.

[85] Sam Peltzman. The effects of automobile safety regulation. *Journal of Political Economy*, 83(4):677–725, 1975.

[86] W. Kip Viscusi. Consumer behavior and the safety effects of product safety regulation. *The Journal of Law and Economics*, 28(3):527–553, 1985.

[87] Manh Hong Duong and The Anh Han. Cost efficiency of institutional incentives for promoting cooperation in finite populations. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 477(2254):20210568, 2021.

[88] Dennis F. Thompson. Moral responsibility of public officials: The problem of many hands. *American Political Science Review*, 74(4):905–916, 1980.

[89] Harold Triedman, Rishi Jha, and Vitaly Shmatikov. Multi-agent systems execute arbitrary malicious code. *arXiv preprint arXiv:2503.12188*, 2025.

[90] European Parliament and Council of the European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act). Official Journal of the European Union, L series, 12 July, 2024.

[91] European Parliament and Council of the European Union. Directive (EU) 2024/2853 on liability for defective products and repealing council directive 85/374/EEC. Official Journal of the European Union, L series, 18 November, 2024.

[92] John W. Meyer and Brian Rowan. Institutionalized organizations: Formal structure as myth and ceremony. *American Journal of Sociology*, 83(2):340–363, 1977.

[93] Richard Ren, Steven Basart, Adam Khoja, Alice Gatti, Long Phan, Xuwang Yin, Mantas Mazeika, Alexander Pan, Gabriel Mukobi, Ryan H. Kim, Stephen Fitz, and Dan Hendrycks. Safetywashing: Do AI safety benchmarks actually measure safety progress? In *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 2024. arXiv:2407.21792.

[94] R. G. Lipsey and Kelvin Lancaster. The general theory of second best. *The Review of Economic Studies*, 24(1):11–32, 1956.

[95] Steven Shavell. Minimum asset requirements and compulsory liability insurance as solutions to the judgment-proof problem. *The RAND Journal of Economics*, 36(1):63–77, 2005.